FluidiTy

Particle–grid multiphysics on one sparse staggered grid

Numerical method and acceptance contracts · 7 November 2026

Companion to the English design specification, 2026-11-07

Abstract. This method note describes a proposed simulation architecture for incompressible liquids, viscous liquids, granular media, snow, elastic solids with cutting, and low-Mach reacting gases. All continua share **one** staggered marker-and-cell (MAC) velocity representation with a candidate sparse adaptive hierarchy: every velocity lives on faces, every scalar at cell centres, and there are no nodal velocities. Particles carry transport and constitutive history; the grid evaluates stress divergence, contact, viscosity, and a variational pressure projection. We state the fourteen-step sub-step, the affine transfer (APIC storing C, per-component deposition and gather moments, full-stencil ghost gathering, ASFLIP), the integrated finite-volume pressure equation with moving walls, ghost-fluid free surfaces and two-phase series coefficients, the viscous and capillary operators, the narrow band with an Eulerian interior, particle split and merge, the reduced low-Mach gas model, the tracked surface and whitewater stages, and the persistent state, memory, transaction, determinism and checkpoint contracts. This is a design synthesis: no production solver or CUDA benchmark has been executed, and no simulation validation result is claimed.

1 Scope, terminology, and conventions

A **continuum** selects a set of governing equations and a particle layout: `Liquid`, `Granular`, `Snow`, `Elastic`, or `Gas`. A **substance** supplies parameter values (water, oil, honey, dry sand, powder snow, rubber, smoke). A **phase** identifies an immiscible region (water and oil are two phases of `Liquid`), whereas a **body** is a connected component for contact: a solid object, one side of a cut, or a Blender object. Two substances may mix within one phase.

The first end-to-end deliverable is a breaking wave with spray, foam and bubbles: an 8 × 20 × 8 m domain, 3.5 cm cells near the camera, 240 frames in at most three hours including all stages and I/O on a 16 GB RTX 5060 Ti with adaptivity (at most six hours tolerated temporarily without it). The reference scene is fixed: 3 m depth at the inlet, then a sloping beach to 0 m over 20 m; second-order Stokes swell with $H \approx 1$ m and $T \approx 4$ s generated by the Ocean Boundary (the paddle is a test variant); a fixed camera 2 m from the breaking point, 35 mm, 1920 px; 24 fps. The host is a Ryzen 7 9700X (8 cores), 64 GB DDR5 and 3 TB of disk; Mesh uses about six CPU cores in parallel with Simulate. Viscous and multi-fluid flow, sand and snow, elastic solids with cutting, then low-Mach gas follow in that delivery order. The architecture is sized for all of them from the start; implementation follows the order.

Sign conventions (used everywhere): $\varphi < 0$ inside the liquid, or inside phase a; $n = \nabla\varphi/|\nabla\varphi|$ is the outward normal; $\kappa = \nabla \cdot n$, so $\kappa = 2/R > 0$ for a spherical drop; $\hat{e}_{\mathrm{up}} = -g/|g|$. Let x denote position, t time, u Eulerian velocity, ρ density, and b body acceleration. Continuum balances are

$$D\rho/Dt + \rho\nabla \cdot u = 0, \quad \rho Du/Dt = \nabla \cdot \sigma_C + \rho b, \quad D/Dt = \partial_t + u \cdot \nabla. \tag{1}$$

σ_C denotes Cauchy stress, plain σ the surface-tension coefficient of a phase pair, γ the heat-capacity ratio of a gas, and $\dot{\gamma}$ a shear rate. At an interface between phases a and b, n points from a to b and jumps are $[q] = q_b - q_a$. $\Delta x_\ell = \Delta x_0 2^\ell$ is the cell size at level ℓ. Indices: p particles, i, j cells or faces, a velocity components, f faces, c children or source cells. V_0 is a reference volume, V a current volume, a_f a physical open face area (m²), and $A_f \in [0, 1]$ a dimensionless obstacle aperture.

For liquids $\sigma_C = -pI + 2\mu D(u)$. For solids σ_C follows an elastic energy and plastic history. For low-Mach gas the thermodynamic pressure p_0 is uniform per pressure component and the projected divergence need not vanish. Whitewater is a one-way visual dispersed phase.

This paper distinguishes owner requirements, provisional numerical choices, acceptance obligations and measured evidence (§ 15.3). Equations below fix sign, volume and unit conventions; changing them requires changing implementation and validation together.

2 One staggered grid for every continuum

2.1 Transport, storage, and spatial unknowns

The material point method (MPM) is a hybrid Lagrangian–Eulerian discretisation. Material points move with the body and carry mass, reference volume, velocity and deformation history; a grid receives their contributions, evaluates forces and contact, and returns velocity. Scratch transfer accumulators reset each sub-step; particle history and persistent Eulerian velocity, occupancy, pressure and gas inventories retain their declared ownership [1, 3]. A MAC grid is a staggered placement of unknowns: pressure at cell centres and each normal velocity component at the centre of the faces perpendicular to its axis, which pairs face pressure gradients with cell flux divergence and avoids the collocated checkerboard.

2.2 One velocity representation

FluidiTy places **all** velocities of **all** continua on the faces of a single MAC grid, one component a per face normal to a. Pressure p, signed distance φ, phase fractions and gas scalars live at cell centres. There is no nodal velocity anywhere. Solids use MPM on a staggered grid (Stomakhin et al. 2014 [18]); affine transfers use MAC-APIC (Ding, Shinar & Schroeder 2020 [2]). Each component has its own quadratic B-spline support on its own shifted lattice.

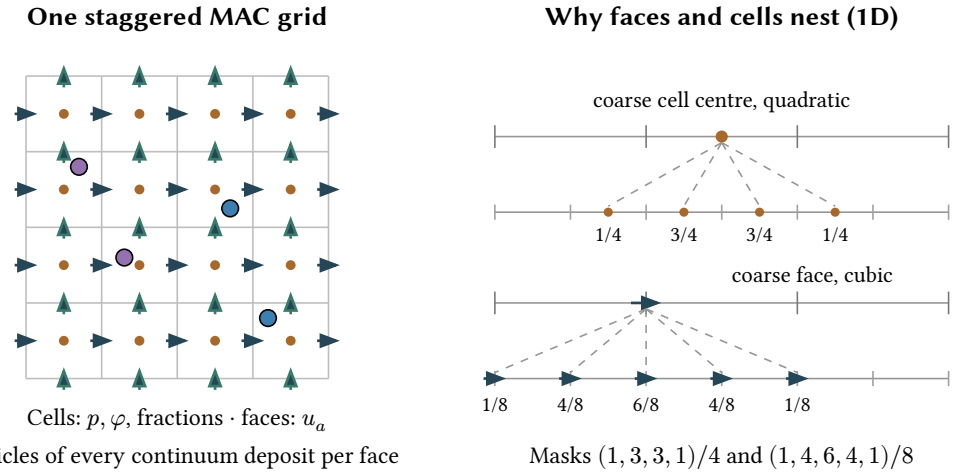


Figure 1: Left: one staggered grid. Cells store p, φ and fractions; faces store one normal velocity component; particles of every continuum deposit into the face lattices. Right: in 1D, a coarse quadratic B-spline whose degrees of freedom sit at cell centres is an exact combination of four fine ones (mask $(1,3,3,1)/4$), and a coarse cubic B-spline centred on a face is an exact combination of five fine ones (mask $(1,4,6,4,1)/8$).

Quadratic splines centred on corners or faces do not nest under dyadic refinement.

The owner requirement is a single MAC velocity representation (D3). The following arguments motivate its candidate adaptive realisation. **(i) Nesting.** Truncated hierarchical B-splines (THB) at 2:1 transitions in this candidate require the coarse basis to be spanned by fine functions. A quadratic B-spline is nested by dyadic refinement only when its degrees of freedom sit at cell centres. On MAC faces, the transverse components are cell-centred along their transverse axes and stay quadratic; the normal component is face-centred along its own axis and becomes a **cubic** B-spline in the M3 candidate, which nests. Then $D_{pa} = \mathrm{diag}(\Delta x^2/3 \text{ on } a, \Delta x^2/4 \text{ elsewhere})$, against $\Delta x^2/4 I$ for the uniform quadratic M1–M2 kernel. A bounded THB experiment must confirm the actual nesting and patch identities after the uniform baseline; storage is identical in both cases. **(ii) No dual representation.** Maintaining collocated solid velocities beside MAC fluid velocities would require a second representation and a conservative face-to-node coupling; the single-grid architecture avoids that obligation. **(iii) Native boundary treatment.** Ghost-fluid free surfaces, obstacle apertures and integrated fluxes are face quantities. Voronoi or power cells (no tensor-product B-splines, high cost) and a Rhie–Chow-stabilised collocated grid (checkerboard modes, no native ghost fluid) are rejected.

The simulation interface (`sdf_sim`, fractions, θ, κ) is owned by Simulate. A rendered surface is reconstructed afterwards by a separate stage at a different resolution; output smoothing is never a physical capillary model, and Simulate never reads it.

2.3 Candidate sparse hierarchy and size control

In the candidate layout a **block** holds 4^3 cells and owns its 64 cells and the 3×64 low-side faces; high-side faces are read from neighbours through the block table. For component a and axis d, samples sit at $(k + s_{ad})\Delta x$ with $s_{ad} = 0$ if $d = a$ and $1/2$ otherwise. Quadratic support starts at $b_d = \lfloor x_d/\Delta x - s_{ad} - 0.5 \rfloor$; cubic normal support starts at $b_d = \lfloor x_d/\Delta x \rfloor - 1$. Particle bins use the common half-offset rule $B_d = \lfloor (x_d/\Delta x - 1.5)/4 \rfloor$, independent of component. A bin owns a shared 8^3 tile with indices $4B_d, \ldots, 4B_d + 7$. For displacement at most $0.5\Delta x$ this contains every uniform quadratic and cubic shifted support. THB requires checking its actual expanded support separately. Every contribution checks tile inclusion before writing; an overflow uses the global path and requests RETRY_DT if a required block is absent. The target displacement CFL is $0.45\Delta x$, based on the actual advecting velocity. An open-addressed GPU hash table maps 64-bit keys (4-bit level and three signed 20-bit offset coordinates, with reserved empty and tombstone values) to dense block indices. Neighbouring blocks differ by at most one level. A 4:1 ratio is rejected: it would split a particle into 64, make one coarse face face 16 fine faces, and THB is only guaranteed under gentle grading. The cited sources do not validate this 3D GPU THB-MPM MAC combination; this is the first project risk, with a go/no-go after an independent uniform reference and real-GPU baseline. Adaptivity is a firm objective (D9): if THB nesting or the patch test fails, an adaptive alternative on boxes (GIMP-style transitions as in AG-MPM, Gao et al. 2017) is tried first, again with a patch test; meanwhile M3 delivers "uniform sparse + narrow band" (≈ 6.8 GB) with FC1 relaxed to at most six hours, and adaptivity stays an open milestone rather than being abandoned.

The requested cell size is an upper bound $h_{\text{req}} = \min\left(h_{\text{phys}}, h_{\text{interface}}, h_{\text{support}}, h_{\text{camera}}, h_{\max}\right)$ with $h_{\text{camera}} \leq \sqrt{8\ \text{err}/|\kappa|}$ and $\text{err} = \text{pixel size} \cdot \text{depth}/\text{focal length}$. Refinement is immediate when $\Delta x > h_{\text{req}}$. Eight sibling blocks coarsen only if $2\Delta x \leq 0.8 h_{\text{req}}$ on the parent and its halo (support plus CFL displacement) **uninterrupted** for a hysteresis time T_{hyst} of accepted physical time; any veto resets the counter. Vetoes are unresolved interfaces or cavities, sheets or gaps thinner than $4\Delta x$, CPIC bands (the finest level is forced within $2\Delta x_{\text{coarse}}$ of blades), body contact, and singular moments. The camera may request detail, never remove it below the physical minimum. A refused coarsening does not reject the step.

3 The sub-step

Table 1 lists the fourteen steps of the common numerical contract in order; Figures 11a–11c expand them. The ordering fixes the time stage of every transfer, force, boundary condition and state update.

Step	Name	Content and ordering contract
①	Emit	Particle and fuel sources (initial volume, inflow) with frozen Δt; IDs from a scan on (parent id, ordinal).
②	Sort	Stable counting sort per block; fused into the previous G2P2G only when § 14.1 allows it.
③	P2G	After activation and reset, per component a, over structurally allocated compatible pairs $\mathcal{C}_p$ with renormalised N', centred r' and D': stress ($\tau = 0$ for Liquid and Gas), mass, pure momentum and force in separate i64 accumulators. Mass validity is decided in ④.
④	Snapshot	Face validity from deposited mass; Eulerian fallback and overlap blending; $u^0 = \frac{q}{m}$ where valid; save $v_{\text{old}} = u^0$ before any force.
⑤	Forces and thermo	Apply stress, gravity and external fields after the snapshot; Gas receives residual buoyancy and conservative thermodynamic updates, without a second full gravity.
⑥	Waves	Ocean Boundary relaxation and absorbing beach on u (after the snapshot).
⑦	Contact	MPM bodies: per-body face velocities and Coulomb friction. Liquid and Gas: no-penetration is imposed by the projection.
⑧	Interface	Geometry at x^n: `sdf_sim`, fractions, apertures, θ, κ, β_f, J_f, s_{corr}.
⑨	Viscosity	One variational grid operator over band and interior; explicit only with a certified spectral bound, otherwise implicit; fractions from ⑧; before projection.
⑩	Project	Compatibility and gauge only for noncompliant Neumann components; solve and correct with identical k_f, J_f. Last physical write to u before ghost construction and G2P.
⑪	G2P	Frozen complete basis $(x^n, N^n, \nabla N^n)$ with old/new ghost samples: δ before overwrite; affine fit C using D^{full}; true gradient L; deformation and ReturnMap; J, β, v; contact on stored and advecting velocities; continuous position sweep.
⑫	Accumulate	Whitewater birth events per source cell and accepted sub-step, owned by the trial.
⑬	Validate	Finite, admissible, true pressure residual and divergence relation → ACCEPTED, else RETRY_DT, CAPACITY or FATAL.
⑭	Adapt	Scheduled refine, coarsen, split, merge and band reseed after acceptance, in a separate transaction.

Table 1: The sub-step (English design specification). Δt is chosen by the CFL rules of § 5.6, reduced on the GPU.

Taking the snapshot in ④ before gravity, stress, buoyancy, wave relaxation and contact fixes the force increment independently of its splitting. With consistent particle/old-grid history, the unconstrained free-fall oracle requires acceleration g within 0.1 % at every energy setting. Interface geometry in ⑧ is evaluated at x^n from the simulation's own `sdf_sim`; the implicit viscosity in ⑨ needs its fractions and precedes the projection, so the velocity read by G2P is the projected one.

Figures 11a–11c at the end of this note expand Table 1. Arrows cannot express every dependency; the contracts that matter most are: the FLIP snapshot precedes all forces (④); interface geometry precedes viscosity, which precedes the projection (⑧, ⑨, ⑩); the projection is the last write to u; inside G2P, δ is computed before any overwrite, ReturnMap precedes β_p, particle contact acts on v_p before x_p moves, and α appears once in the position update; force and deformation pair outside cut bands, with their discrepancy and boundary work recorded inside them; whitewater events belong to the trial; acceptance advances the clock once; adaptation follows acceptance.

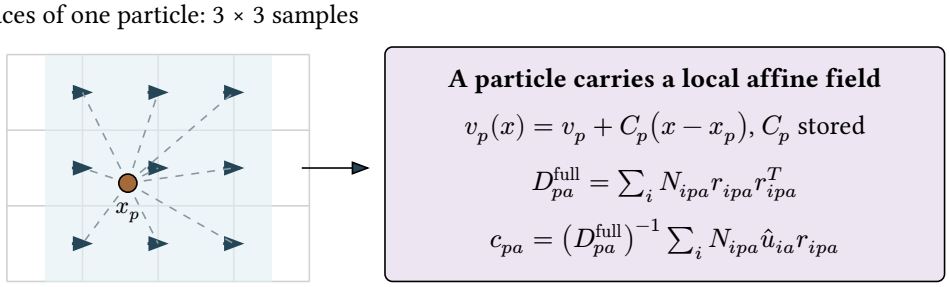
4 Affine particle–grid transfer on the MAC grid

4.1 Stored state and moments

A PIC particle stores v_p only. APIC stores the affine model $v_p + C_{p(x-x_p)}$; C has units s^{-1} and is an affine fit, generally distinct from the true gradient L. Storing C rather than the support-dependent moment $B = CD$ avoids an artificial scale change when support changes, but does not preserve affine energy or angular momentum automatically (§ 9.5). Let $r_{ipa} = x_{ia} - x_p$, with x_{ia} the basis's Greville abscissa. Deposition uses the compatible centred moment D'_{pa} of § 4.4; gathering uses

$$D_{pa}^{\text{full}} = \sum_{i \in \mathcal{S}_{pa}} N_{ipa} r_{ipa} r_{ipa}^T. \tag{2}$$

$\mathcal{S}_{pa}$ is the complete original support, including ghost samples. Without a cut, uniform quadratics give $D^{\text{full}} = \Delta \frac{x^2}{4} I$; transitions require the actual full matrix. If either required moment is ill-conditioned or supporting mass insufficient, local PIC is counted and `kAffineOff` is set. The slot `C` then carries the diagnostic true gradient L; every affine consumer accesses zero through a typed accessor. Split and merge refuse flagged particles.



Full-stencil G2P includes ghost faces; C_{affine} is a fit, while $L = \sum \hat{u} \nabla N^T$ evolves deformation.

Figure 2: Affine reconstruction on the shifted face lattices. Deposition uses compatible samples; gathering completes the original basis with ghosts and uses its own moment.

4.2 P2G, grid update, G2P

For the compatible deposition support, with $r' = r - \mu$ and weights of § 4.4,

$$m_{ia} = \sum_p N'_{ipa} m_p, \quad q_{ia} = \sum_p N'_{ipa} m_{p\left(v_{pa} + c_{pa} \cdot r'_{ipa}\right)}, \quad f_{ia} = -\sum_p V_{0p} \left(\tau_p \nabla N'_{ia}(x_p)\right)_a. \tag{3}$$

$$v_{\text{old},ia} = \frac{q_{ia}}{m_{ia}}, \quad u_{ia} = v_{\text{old},ia} + \Delta t \left(\frac{f_{ia}}{m_{ia}} + g_{K,a} + f_{\text{fields},a} \right). \tag{4}$$

Here $g_K = 0$ for Gas, whose buoyancy is the residual relative to ambient hydrostatic pressure. Activation and reset precede deposition; structural allocation does not depend on the yet unknown mass. Mass, momentum and force are quantised separately before any sum (§ 14.1). The old-grid snapshot is taken after band blending and before every force.

For gathering define complete sample arrays $\hat{u}^n$ and $\hat{u}^{n+1}$ on the frozen original support. Compatible valid samples read the pre-force snapshot and projected final velocity respectively. Missing or incompatible samples are extrapolated separately from these two time stages on the particle's own side; prescribed blade-normal values use the blade velocity at the matching time stage. The old array must never be reconstructed from the post-force field. Affine reproduction requires affine-complete extrapolation of a field compatible with the boundary. Arbitrary extrapolation followed by normal replacement does not establish it. Then

$$v_{\text{pic},a} = \sum_i N_{ipa} \hat{u}_{ia}^{n+1}, \quad \delta_a = v_{pa}^n - \sum_i N_{ipa} \hat{u}_{ia}^n, \tag{5}$$

$$c_{pa} = \sum_i N_{ipa} \hat{u}_{ia}^{n+1} r_{ipa}^T \left(D_{pa}^{\text{full}}\right)^{-1}, \quad L_p = \sum_a e_a \sum_i \hat{u}_{ia}^{n+1} \nabla N_{ia}(x_p)^T, \tag{6}$$

$$v_p = v_{\text{pic}} + \alpha\delta, \quad w_p = v_{\text{pic}} + \beta_p \alpha \delta, \quad x_p \leftarrow x_p + \Delta t w_p. \tag{7}$$

If the support touches an Eulerian fallback face, set $\delta = 0$ explicitly. Within ⑪, compute δ before overwriting the particle velocity; gather C and L separately; form $F_{\text{trial}} = (I + \Delta t L)F$; reject nonpositive determinant; apply ReturnMap; evolve Liquid and Gas $J \leftarrow J \exp(\Delta t \operatorname{tr} L)$; choose β from the post-return state; then

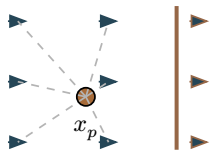
constrain both v_p and w_p by contact and continuously sweep the trajectory (§ 5.5). α occurs once in each update [4].

The conservation properties derived for MAC-APIC [2] depend on its support and position convention. They do not extend automatically to ghost replacement, pressure projection, adaptive resampling or ASFLIP. APIC dissipates some modes. No-force FLIP tests must use both old and new ghost arrays through this executable sequence.

4.3 Energy slider and separation

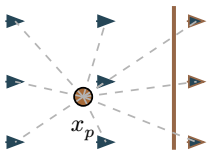
The artist sees one control, **energy** $e \in [0, 1]$ (default 0.6), with $\alpha = 0.99e$ and $\beta_{\max} = e$ (design choice). The separation coefficient is per particle, $\beta_p \in \{0, \beta_{\max}\}$, following Fei et al. [4, §5.3] (the exact criterion is to be re-checked against the source). Liquid particles separate when $\operatorname{tr} L_p > 0$ and $\varphi > -2\Delta x$ (free-surface band), and never near an obstacle. Granular and Snow particles separate when $\det F_E \geq 1$ after the plastic return and out of contact. The initial Gas baseline uses APIC with $\alpha = \beta = 0$, without liquid free-surface separation; later gas blending has a separate transport/effect gate. **Elastic particles always have $\beta_p = 0$**: a non-zero β displaces an elastic body without generating stress. The dam-break look-book must show a useful, measured change with e while retaining force-response and stability gates; no universal monotonic-energy theorem is assumed.

P2G · compatible, normalised stencil



$$N' = \tfrac{N}{S};\ r' = r - \mu;\ D' = \textstyle\sum N' r' r'^T$$
$$q_{ia} \mathrel{+}= N' m_{p(v_{pa} + c_{pa} \cdot r')}$$
$$f_{ia} \mathrel{+}= -V_{0p} \left(\tau_p \nabla N'_{ia}\right)_a$$

G2P · full stencil with ghosts



$$v_{\text{pic}} = \textstyle\sum N \hat{u}^{\text{new}};\ D^{\text{full}} = \sum N r r^T$$
$$\delta = v_p - \textstyle\sum N \hat{u}^{\text{old}};\ L = \sum \hat{u}^{\text{new}} \nabla N^T$$

Blade-normal ghost component = blade velocity

The cut-band work defect and blade impulse are counted; outside cuts the true-gradient work identity holds.

Figure 3: Compatible centred deposition and complete old/new ghost gathering have distinct weights and moments. Exact stress–deformation work pairing applies outside cut bands; cut replacement work and reactions are recorded.

4.4 Renormalised masks for cutting and inactive faces

CPIC [3] excludes particle–face pairs across a blade. Structurally unavailable faces are excluded from deposition as well; mass validity is determined afterwards. Per component, over the compatible set $\mathcal{C}_p$,

$$S_p = \sum_{i \in \mathcal{C}_p} N_{ip}, \quad N'_{ip} = \frac{N_{ip}}{S_p}, \quad \nabla N'_{ip} = \frac{\nabla N_{ip} - N'_{ip} \sum_{j \in \mathcal{C}_p} \nabla N_{jp}}{S_p}, \tag{8}$$

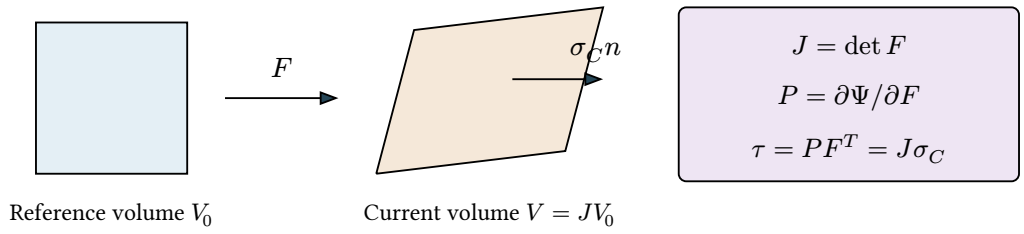
$$\mu_p = \sum_i N'_{ip} r_{ip}, \quad r'_{ip} = r_{ip} - \mu_p, \quad D'_p = \sum_i N'_{ip} r'_{ip} r'^T_{ip}. \tag{9}$$

Thus $\sum N' = 1$ and $\sum \nabla N' = 0$: centred deposition preserves particle momentum and gives zero net internal force for constant stress. Gathering instead uses the complete $N, \nabla N, r, D^{\text{full}}$ above. If $S_p < 0.1$ or D' is singular, use counted free flight/local PIC, suppress separation and affine deposition, and constrain both velocities before advection. Skipping the gather must not skip the position update.

5 Stress, contact, and constitutive models

5.1 Stress measures and the work identity

Let $F = \partial \frac{x}{\partial} X$, $J = \det F$, and $V = JV_0$. For energy Ψ per reference volume, $P = \partial \frac{\Psi}{\partial} F$ and $\tau = PF^T = J\sigma_C$. Solids return τ for force deposition $f_{ia} = -V_{0p} \left(\tau_p \nabla N'_{ia}\right)_a$; an extra J would count volume change twice. Liquid and Gas return zero particle stress: viscosity is a grid operator.



$f_{ia} = -\sum_p V_{0p}\big(\tau_p \nabla N'_{ia}(x_p)\big)_a$; outside cuts $N' = N$ and force–deformation work is exact.

Figure 4: Stress measures and volume conventions. Kirchhoff stress rescales Cauchy stress by J; deposition and deformation pair on complete unmasked supports.

Outside cuts, force and deformation use the same true basis gradient, giving $\sum_i u_i \cdot f_i = -\sum_p V_{0p}\tau_p : L_p$. The MLS replacement $ND^{-1}r$ is not used for deformation or stress force. On a cut support define L'_p from compatible $\nabla N'$ and real grid samples; the force work is $-\sum V_0\tau : L'$, while deformation uses the complete ghost L. Record the work discrepancy $\Delta W_{\text{cut}} = \Delta t \sum V_0\tau : (L - L')$, the impulse and torque introduced by boundary replacement, and blade work. Count each velocity-replacement reaction once; never convert an arbitrary work residual into an impulse. Equal and opposite boundary impulses and torques enter the owning body; a prescribed blade's external actuator ledger receives its reaction. This records the cut approximation rather than asserting exact internal work pairing there. Ordinary C is never substituted for L.

5.2 Granular media: Hencky elasticity and Drucker–Prager return

State: $x, v, C, m, V_0, F_E, \log J_{\text{pl}}$, body. With $F_E = U\Sigma V^T$ and $\varepsilon = \log\Sigma$, $\tau = U(2\mu\varepsilon + \lambda\,\mathrm{tr}(\varepsilon)I)U^T$. The return of Klár et al. [7] acts on the trial singular values of $F_{\text{trial}} = (I + \Delta t L)F_E$:

$$\hat{\varepsilon} = \varepsilon - (\mathrm{tr}\,\varepsilon/3)I, \quad \alpha_{\text{DP}} = \sqrt{2/3}\big(2\sin\varphi_f\big)/\big(3 - \sin\varphi_f\big), \quad \delta\gamma = \|\hat{\varepsilon}\| + (3\lambda + 2\mu)/(2\mu)\,\mathrm{tr}(\varepsilon)\alpha_{\text{DP}} \tag{10}$$

Use the translated-cone coordinate $t = \mathrm{tr}\,\varepsilon - c_{\text{coh}}$. The apex branch $t \geq 0$ precedes any norm division: set $\varepsilon = \big(\frac{c_{\text{coh}}}{3}\big)I$ (zero for cohesionless sand). Otherwise use $\delta\gamma = \|\hat{\varepsilon}\| + (3\lambda + 2\mu)t\frac{\alpha_{\text{DP}}}{2\mu}$. If $\delta\gamma \leq 0$ the trial is elastic; otherwise $\varepsilon \leftarrow \varepsilon - \delta\gamma\frac{\hat{\varepsilon}}{\|\hat{\varepsilon}\|}$. The apex branch handles zero deviatoric norm without division. Reconstruct $F_E = U\exp(\varepsilon)V^T$ and update $\log J_{\text{pl}}$ by the trial-minus-return log determinant. Cohesion here is a dimensionless translated logarithmic-strain cone parameter, requiring material calibration. Artist controls are friction angle and cohesion; tests include the pile angle, zero tensile stress for cohesionless material, and the cohesive isotropic apex.

5.3 Snow

Following Stomakhin et al. [19], $F_E = R_E S_E$, $J_E = \det F_E$, $J_{\text{pl}} = \exp\big(\log J_{\text{pl}}\big)$ and

$$\tau = 2\mu(F_E - R_E)F_E^T + \lambda(J_E - 1)J_E I, \quad \mu = \mu_0 e^{\xi(1-J_{\text{pl}})}, \quad \lambda = \lambda_0 e^{\xi(1-J_{\text{pl}})}. \tag{11}$$

The trial $F_{\text{trial}} = U\Sigma V^T$ is clamped to $\Sigma_c = \mathrm{clamp}(\Sigma, 1 - \theta_c, 1 + \theta_s)$, $F_E = U\Sigma_c V^T$, and $J_{\text{pl}} \leftarrow J_{\text{pl}}\det\Sigma/\det\Sigma_c$, which keeps $\det F_E J_{\text{pl}} = J_{\text{total}}$ (`snow_jp_invariant`). One artist control, compactness, drives ξ, θ_c, θ_s and E_0.

5.4 Elastic solids and cutting

Compressible neo-Hookean (default) and fixed-corotated (option, stable under large rotations):

$$\tau = \mu(FF^T - I) + \lambda\ln J I, \quad \tau = 2\mu(F - R)F^T + \lambda(J - 1)JI, \tag{12}$$

with $\mu = E/(2(1+\nu))$ and $\lambda = E\nu/((1+\nu)(1-2\nu))$; here μ is a shear modulus, not a viscosity. Elastic models require $J > 0$. Controls are stiffness E and bounce (Rayleigh damping on L); $\beta_p = 0$. For cutting, each blade k defines an unsigned distance d_k and band $B_k = \{d_k < r_{\text{cut}}\}$. A face carries $\text{affinity}_k = [i \in B_k]$ and side_k; a particle memorises $\text{side}_{k(p)}$ when it enters B_k (persistent, after BladeMaster). A pair is compatible iff for all k, $\text{affinity}_{k(i)} \wedge \text{affinity}_{k(p)} \Rightarrow \text{side}_{k(i)} = \text{side}_{k(p)}$. Incompatible pairs leave $\mathcal{C}_p$; in G2P the blade-normal component takes the blade velocity. The two sides of a cut become two bodies. Damage fracture (CD-MPM) is deferred.

5.5 Contact, boundaries, and rigid bodies

Blender obstacles become solver-side SDFs with sampled velocity. Fluids see them **only** through apertures and u_{obst} in the projection (§ 6); MPM bodies see them through a face contact rule: with $v_{\text{rel}} = v - v_{\text{obst}}$ and

$v_n = v_{\text{rel}} \cdot n < 0$, retain the incoming v_n for the friction limit, set the outgoing normal component to zero, and use $v_t \leftarrow \max(0, |v_t| - \mu|v_n|)\hat{t}$, with the tangential part reconstructed from the four neighbouring faces. Store the incoming normal velocity before zeroing the outgoing part; Coulomb friction uses that incoming impulse. One artist control, adhesion μ, selects sticky, slipping or separable behaviour. In G2P, near-contact inward velocity is removed from both stored v_p and actual advecting w_p; active contact sets $\beta_p = 0$. Sweep the relative trajectory against the time-dependent obstacle to its first contact, apply the rule there, then advance the remaining time subject to renewed contact. Use conservative advancement with certified SDF distance/Lipschitz and motion bounds, or continuous swept-triangle intersection for the represented animated geometry. A fixed grid-relative sample spacing cannot guarantee detection of a thin wall. Final projection out of an obstacle repairs numerical penetration but is not collision detection. Velocity changes contribute $m_p \Delta v$, matching torque and moving-wall work to the owning obstacle or actuator ledger. This also constrains inward velocity from the FLIP residual.

Faces touched by several simulated bodies (water on sand, water against rubber) carry one velocity per body ($K \leq 4$, design choice) in a sparse table, merged or separated by Coulomb with normals from mass gradients (Nairn; DC-APIC; IQ-MPM). Exceeding K raises E-CONTACT-BODY-CAPACITY rather than gluing bodies together.

Fluid against simulated MPM bodies (design choice, partitioned). In the projection, the face fraction occupied by an MPM body is treated as an obstacle: $A_f = 1 - \text{solid fraction}_f$ and $u_{\text{obst},f}$ is the body face velocity after ⑦. The resulting pressure acts on the body at the next sub-step, $f_{ia} \mathrel{+}= p_f\big(|f| - a_f\big)n_f$ distributed over the body faces, with n_f outward from the fluid. Overlapping blockers require their geometric union and unambiguous wall motion, or a counted veto; summing overlapping solid fractions does not give exact union geometry. The one-sub-step lag is a known added-mass risk for light bodies; it is monitored by a floating-elastic-cube test and a logged added-mass ratio, with a monolithic coupling (IQ-MPM) as fallback.

A wave paddle is an animated obstacle: it pushes water through the moving-wall flux of the projection. The Ocean Boundary applies, in ⑥ after the snapshot, $u \leftarrow u + \big(1 - e^{-\Delta t/\tau(\xi)}\big)\big(u_{\text{target}} - u\big)$ with $\tau(\xi) = \tau_0/w(\xi)$, $w(\xi) = \big(e^{\xi^{3.5}} - 1\big)/(e-1)$ (Jacobsen et al. [20]); the water level is driven toward η_{target} by emitting or deleting particles, and second-order Stokes waves subtract the return current. An absorbing beach uses the same form with a rest target.

Two-way rigid bodies (decided, M6). In v1 a Blender object is a one-way obstacle; two-way coupling arrives with solids at M6. With n_f outward from the body, $F_B = -\sum_f p_f\big(|f| - a_f\big)n_f + \sum \text{contact impulses}/\Delta t + m_B g$ plus the matching torque, integrated by Newton–Euler, and the body is reinjected as an obstacle with $u_{\text{obst}} = v_B + \omega_B \times (x - x_B)$. The coupling is partitioned and explicit; added-mass ratios are monitored, with sub-iterations if needed. The acceptance test is a 500 kg/m³ cube floating at half height within ±2 %.

The rigid-body table stores positive mass, body-frame inertia, centre, unit quaternion, linear/angular velocity and force/torque accumulators; pose/history is checkpointed. With $I_{\text{world}} = R I_{\text{body}} R^T$, integrate

$$\dot{v} = \frac{F}{m}, \quad \dot{\omega} = I_{\text{world}}^{-1}(T - \omega \times (I_{\text{world}}\omega)), \tag{13}$$

and orientation consistently with world angular velocity while preserving quaternion normalisation. Pressure and contact reactions enter the torque once. Rotation/sweep bounds use this same pose evolution; translation alone is not a complete rigid-body update.

5.6 Time step

The advective target is $\Delta t \leq 0.45\Delta \frac{x_{\min}}{v_{\max}}$, with $v_{\max}$ bounding the actual particle w_p, Eulerian band faces, source velocities and obstacle motion. Sources and walls are evaluated before emission without consuming IDs; the particle estimate comes from the previous step and is checked a posteriori. The shared-tile inclusion limit is $0.5\Delta x$ for uniform kernels; actual THB supports are checked individually. To reach a frame boundary, choose $n = \left\lceil \frac{t_{\text{frame}} - t}{\Delta} t_{\max} \right\rceil$ and $\Delta t = \frac{t_{\text{frame}} - t}{n}$, avoiding tiny residual steps.

Solids require $\Delta t \leq C_s \Delta \frac{x}{c_K}$, where `K::WaveSpeed` certifies a bound on the largest eigenvalue of the full acoustic tensor over propagation directions in the current returned state, using the reference density and current hardened tangent. Principal-axis Hencky coefficients alone do not bound oblique modes. Directional checks require a certified enclosure of the unsampled directions; an unqualified sampled maximum is insufficient. Loss of ellipticity or a nonpositive acoustic mode is an admissibility failure with an explicit diagnostic, not a negative square root. For a principal frame with positive singular values s_i, let $A_{ijkl} =$

$\partial^2 \frac{\Psi}{\partial F_{ij} \partial F_{kl}}$ and $B_{ijkl} = A_{ijkl} s_j s_l$. Major symmetry makes the flattened B symmetric; the acoustic tensor is its restriction to $a \otimes n$, an isometry for unit n. Thus the conservative bound

$$c_K^2 = \left(\frac{1}{\rho_0}\right) \max_{ij} \sum_{kl} |B_{ijkl}|, \quad \rho_0 = \frac{m}{V_0}, \tag{14}$$

controls every direction. It is a safe upper bound, not an exact wave speed. For Hencky $t_i = 2\mu \ln s_i + \lambda \ln J$, the nonzero blocks are $A_{iiii} = \frac{2\mu+\lambda-t_i}{s_i^2}$, $A_{iijj} = \frac{\lambda}{s_i s_j}$ and, for $i \neq j$,

$$a_{ij} = \frac{t_i - t_j}{s_i^2 - s_j^2}, \quad b_{ij} = \frac{\left(\frac{s_j}{s_i}\right) t_i - \left(\frac{s_i}{s_j}\right) t_j}{s_i^2 - s_j^2}, \tag{15}$$

with shear block $\begin{pmatrix} a_{ij} & b_{ij} \\ b_{ij} & a_{ij} \end{pmatrix}$. At equal stretches use limits $a = \frac{\mu}{s_i^2}$, $b = \frac{\mu - t_i}{s_i^2}$. The row-sum bound follows from these three normal and three shear blocks; use stable divided differences near equality. Other laws use their full consistent tangent or a proved sharper bound. Reject nonfinite or unsupported states, including singular values below the declared floor; never hide loss of ellipticity by clipping a negative squared speed. C_s still accounts for the discrete spatial stencil. Independent directional differentiation checks validate the implementation of the enclosure.

For the corotational and snow energy $\Psi = \mu \sum_{i(s_i-1)}^2 + \frac{\lambda(J-1)^2}{2}$, the normal tangent block is $A_{iiii} = 2\mu + \lambda\left(\frac{J}{s_i}\right)^2$, $A_{iijj} = \lambda(2J-1)\frac{J}{s_i s_j}$. Its shear coefficients are $a_{ij} = 2\mu\left(1 - \frac{1}{s_i+s_j}\right)$, $b_{ij} = 2\frac{\mu}{s_i+s_j} - \lambda(J - 1)\frac{J}{s_i s_j}$. Hardened moduli are frozen during this elastic linearization; plastic updating is separate. These analytic blocks feed the same spatial-tangent row bound, avoiding a dense Hessian per particle.

Explicit viscosity requires $\Delta t \Lambda_\mu \leq 2(1 - \varepsilon_{\text{stab}})$ with a certified assembled spectral upper bound (§ 7.1); otherwise use the implicit branch. Add the capillary bound of § 8 and the cavity bound $\Delta t \leq \frac{0.5}{\omega_{\text{cav}}}$ of § 10.3. GPU reduction selects the bound without a host read. Per-region stepping is deferred.

6 Pressure projection

6.1 Integrated finite-volume form

The projection follows Batty et al. 2007 and Ng et al. 2009 [21, 22]. For each face f oriented $i \to j$, with $|f|$ the geometric face area,

$$a_f = A_f\,|f|, \quad d_f = \text{centre distance } \big(\theta d_f \text{ if cut by the free surface}\big), \quad k_f = \Delta t a_f \beta_f / d_f, \quad \beta_f = 1/\rho_f. \tag{16}$$

The row of cell i is

$$\sum_{j \text{ unknown}} k_f (p_i - p_j) + \sum_{f \text{ Dir}} k_f p_i = b_i, \tag{17}$$

$$\begin{aligned} b_i = V_i s_i &- \sum_f a_f u_f^\star n_{if} \\ &- \sum_f (|f| - a_f) u_{\text{obst},f} n_{if} \\ &- \sum_{j \text{ unknown}} k_f J_f + \sum_{f \text{ Dir}} k_f p_\Gamma. \end{aligned} \tag{18}$$

and the correction uses the same k_f, β_f, J_f:

$$u_f = u_f^\star - \big(\Delta t \beta_f / d_f\big)\big(p_j - p_i - J_f\big). \tag{19}$$

V_i is the fluid volume of the cell and u_{obst} the velocity of the obstacle (animated, deformable, paddle) on the blocked part of the face. Without the moving-wall term the paddle does not push water; `piston_work` requires the displaced volume to match $\int u_{\text{obst}} \cdot n \, \mathrm{d}A \, \mathrm{d}t$ within 1 %. Units: $k_f p$ and every right-hand term are volume fluxes in m³/s. Rows are **not** normalised by V_i.

Free surface (ghost fluid, Gibou et al. [23]): $\theta = \varphi_i / \big(\varphi_i - \varphi_j\big)$, bounded below by 0.05 as a documented regularisation; the face becomes Dirichlet with $p_\Gamma = p_{\text{gas,gauge}} + \sigma\kappa$ (ambient gas gauge pressure zero). On this Dirichlet link set $J_f = 0$: capillarity is already in p_Γ and must not be counted twice. **Same phase**: $\beta_f = 1/\rho$ (miscible: ρ averaged between i and j). **Two immiscible phases** (Kang, Fedkiw & Liu [24]): θ is the fraction of the segment $i \to j$ on the side of i's phase, and continuity of $\beta \partial p / \partial n$ puts the two half-links **in series**:

$$\beta_f = 1/\big(\theta\rho_i + (1-\theta)\rho_j\big) \quad \text{(no bound on } \theta\text{)}. \tag{20}$$

This replaces an arithmetic mean of $1/\rho$, which overestimates the water–air coefficient roughly 250-fold. The oriented capillary jump is

$$[p] = [2\mu n \cdot Dn] - \sigma\kappa, \quad J_f = p_{\text{side } j} - p_{\text{side } i} = \begin{cases} [p] & \text{if } i \in a \\ -[p] & \text{if } i \in b \end{cases} \tag{21}$$

with n from a to b and $[q] = q_b - q_a$; in v1 the viscous term is omitted (§ 7.2), so $J_f = -\sigma\kappa_f$ for $i \in a$. J_f enters the right-hand side as $-k_f J_f$ in row i and $+k_f J_f$ in row j, and Equation 19 with the same sign. For a static drop of a, $p_a - p_b = \sigma\kappa = 2\sigma/R$ (`static_drop_jump_orientation`). At M4 the normal viscous stress jump is included.

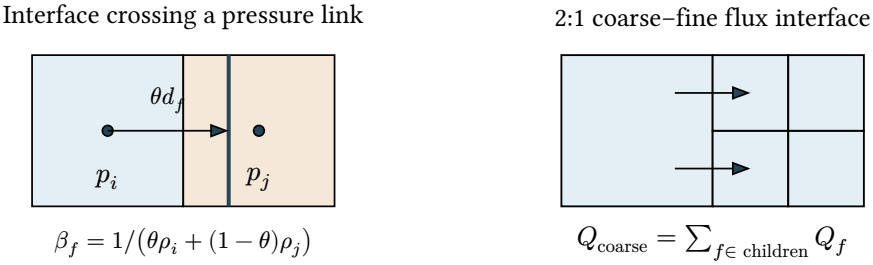


Figure 5: Left: an interface crossing a pressure link at fraction θ; the two half-links are resistances in series. Right: at a 2:1 T-junction the coarse face receives exactly the sum of the fine physical fluxes.

The pressure metric. Orient $R_{if} = +1$ at a link's tail and $R_{jf} = -1$ at its head. Then $D = R\ \mathrm{diag}(a)$ maps face velocity to integrated outward fluid flux. The integrated pressure-gradient operator is $G_{\text{grad}} = -D^T$; the physical pressure force is $-G_{\text{grad}}p = D^T p$. With face inertia

$$M_{ff} = a_f \frac{d_f}{\beta_f} > 0, \quad u^+ = u^\star - \Delta t M^{-1} G_{\text{grad}} p, \tag{22}$$

the correction agrees with Equation 19, including its separate boundary/jump lifting. Closed faces are removed or prescribed. This selects one positive discrete quadrature for pressure and viscous kinetic energy; it is not a claim that arbitrary particle-lumped face masses equal geometric dual masses. The velocity-gradient operator is $M^{-1}G_{\text{grad}}$, with its mass metric explicit. On a tiny aperture, use the same a, d, β in mass, conductance and correction rather than silently replacing the inertia in one branch.

T-junctions and matrix structure. A coarse face receives the sum of its fine physical fluxes. With $k_f = \Delta t a_f \frac{\beta_f}{d_f}$, Dirichlet anchoring R_{Dir} and compliance C, the integrated pressure matrix is

$$A = R\ \mathrm{diag}(k) R^T + R_{\text{Dir}} + C = \Delta t D M^{-1} D^T + R_{\text{Dir}} + C. \tag{23}$$

Positive conductance and anchoring give a positive semidefinite matrix, and positive definiteness requires each constant component to be anchored or gauge-restricted. This is an algebraic property of the assembled operators; correct geometry and adaptive consistency require separate tests. Second-order transition accuracy remains a target [6]. The reserved compliance is $C_{ii} = \frac{V_i}{\rho_i c_i^2 \Delta t}$, with $Cp^\star$ on the right; both vanish in incompressible mode.

6.2 Sources and compatibility

s_i (s^{-1}) separates prescribed physical sources from a numerical Liquid density correction. Define a same-phase cell-centred sensor using declared nonnegative quadrature weights $N_{\text{cell},ip}$,

$$\overline{J}_i = \frac{\sum_p N_{\text{cell},ip} J_p V_{0p}}{\sum_p N_{\text{cell},ip} V_{0p}}, \quad e_i = -\frac{\ln\big(\overline{J}_i\big)}{t_{\text{corr}}}, \quad t_{\text{corr}} > 0. \tag{24}$$

Guard its denominator before evaluation. These are cell weights, distinct from the face-component transfer stencils. In incompressible mode the sensor-supported band uses $s_{\text{corr},i} = \mathrm{clip}(e_i, -s_{\max}, s_{\max})$ and the particle-free core uses zero. Compressed particles have $\overline{J} < 1$, giving positive expansion. The time and rate scales are physical parameters validated across h and Δt.

For a sealed zero-compliance component, project only the numerical band correction to bounded zero mean: choose a scalar λ satisfying

$$\sum_{i\in \text{ band}} V_i\ \mathrm{clip}(e_i - \lambda, -s_{\max}, s_{\max}) = 0, \quad s_{\text{corr},i} = \mathrm{clip}(e_i - \lambda, -s_{\max}, s_{\max}). \tag{25}$$

The monotone scalar solve retains the clamp and leaves core cells at zero; it does not spread a band mean into the core. Physical sources remain separately subject to compatibility. For every component without Dirichlet or positive compliance,

$$\sum_i V_i s_i = \sum_{\text{walls}} (|f| - a_f) u_{\text{obst}} \cdot n. \tag{26}$$

Gas evolves its thermodynamic p_0 as described in § 10. Any incompatible physical remainder raises E-PRESSURE-INCOMPATIBLE before PCG without a time-step retry. Fix one gauge per unanchored non-compliant component. Positive compliance instead preserves physical mean pressure and receives neither a gauge nor a zero-mean correction.

6.3 Interface geometry and solver

Step ⑧ reduces cut cells to fractions: for each interface cell and phase, a volume fraction, apertures, θ, κ, β_f and J_f, from one plane per cell (value and gradient of `sdf_sim`), stored in FP16 for interface cells only. No polygon is built. A single plane cannot encode both sides of a sheet thinner than a cell or a triple junction; those cases are handled by refinement or by the surface stage (§ 11). The solver is PCG preconditioned by matrix-free multigrid on blocks [6, 25], with FP32 vectors and products and FP64 dot products and coarse level. Convergence requires $\|r\| \le \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}}\|b\|$ **and** $\max_i |r_i|/V_i \le \varepsilon_{\text{div}}$ in s^{-1}; with identical assembly and correction the error in the selected incompressible or compliant divergence relation equals $-V^{-1}r$, so an integrated tolerance alone could hide large divergence in tiny cut cells. The true residual is recomputed after the loop. Non-convergence gives RETRY_DT (at most twice) then E-PRESSURE-NOT-CONVERGED. Polyhedral integration of solid stresses is deferred.

7 Viscosity

Dynamic viscosity μ is in Pa·s and $\nu = \mu/\rho$ in m^2/s. The operator is always the tensor form $\nabla \cdot (2\mu D(u))$ with $D = (\nabla u + \nabla u^T)/2$, never $\mu\Delta u$: the reduction needs constant μ and zero divergence, and fails at free surfaces and between viscosities.

7.1 One full-grid variational operator

Step ⑨ applies viscosity over both the particle band and Eulerian interior, using fractions from ⑧. With the same positive face inertia $M_{ff} = a_f \frac{d_f}{\beta_f}$ as the pressure metric and symmetric-gradient operator E,

$$K_\mu = E^T W_\mu E, \quad W_\mu = 2\mu V, \quad b_{\text{wall}} = E^T W_\mu d_{\text{obs}}, \quad u^T K_\mu u \ge 0, \tag{27}$$

where strain is $Eu - d_{\text{obs}}$, the mass and strain quadratures use the same declared geometry, zero-volume degrees of freedom are removed or prescribed, and off-diagonal tensor terms are counted twice, coefficients at centres and edges are volume-averaged, and the free boundary is traction-free [8]. No particle viscosity is deposited, and C is not used as a stress gradient. M1 uses zero viscosity; the full operator is delivered at M4.

For forward Euler $u^+ = u^\star + \Delta t M^{-1}(b_{\text{wall}} - K_\mu u^\star)$, positive semidefinite K_μ alone does not guarantee dissipation. With stationary homogeneous walls,

$$\Delta E = -\Delta t u^T K_\mu u + \left(\Delta \frac{t^2}{2}\right) u^T K_\mu M^{-1} K_\mu u. \tag{28}$$

Define $H = M^{-\frac{1}{2}} K_\mu M^{-\frac{1}{2}}$ on positive-mass unconstrained degrees of freedom. A conservative Gershgorin bound is $\Lambda_\mu = \max_i \left(H_{ii} + \sum_{j \ne i} |H_{ij}|\right)$; safe assembly or a proved row-sum enclosure includes every contribution and rounds upward with a positive safety margin. Select one branch over each whole coupled component, with the explicit branch allowed only if $\Delta t \Lambda_\mu \le 2(1 - \varepsilon_{\text{stab}})$ over all its rows; do not mix independent rowwise material decisions. The diffusion-number shortcut $\nu \Delta \frac{t}{\Delta} x^2 \le \frac{1}{12}$ is allowed only for a separately certified uniform constant-coefficient stencil. Cuts, slivers, coefficient contrast and level transitions require the assembled bound or implicit fallback.

The implicit branch solves $(M + \Delta t K_\mu)u = Mu^\star + \Delta t b_{\text{wall}}$. Both branches account for prescribed wall work; kinetic-energy decay is asserted relative to that work, not absolutely for moving walls. The subsequent projection enforces the selected incompressible or compliant divergence relation. Herschel–Bulkley materials use the same operator with $\mu_{\text{eff}} = \frac{\tau_y + K\dot\gamma^n}{\dot\gamma}$ and $\dot\gamma = \sqrt{2D : D} \ge \dot\gamma_{\min}$.

7.2 Interfaces between viscosities

In v1, viscosity then pressure is a fractional step. It leaves a normal traction error $2[\mu]D_{nn}$ at interfaces between two viscosities, which is accepted and measured (`planar_extension_traction`). Decided (D6): M4 keeps the fractional step first, and moves to a coupled Stokes solve (Larionov et al. [9]) when the predeclared traction or visible-shape gate fails. Before running, freeze analytic or independently converged coupled-Stokes references for planar extension and Couette at viscosity ratios 1, 10 and 1000. Measure area-weighted RMS and maximum normal/tangential traction errors using a nonzero fixture stress scale (for example $2\mu_{\text{ref}}\frac{U}{L}$) and an absolute floor. The initial gate is 5 % RMS plus a fixture-declared maximum bound at two resolutions and reduced Δt; thresholds cannot be tuned after seeing results. That saddle-point system is indefinite and needs a Schur complement or block preconditioner rather than plain PCG.

8 Surface tension

In v1 capillarity is the jump $\sigma\kappa$ inside the projection, with $\kappa = \nabla\cdot(\nabla\varphi/|\nabla\varphi|)$ computed on `sdf_sim` **at every sub-step**, at the same time as θ. The tracked mesh is never read by Simulate (`curvature_no_mesh_dependency`). Applying both an area force and the full Laplace jump would double-count capillarity; pressure gradients and capillary jumps must balance in the same discrete operator so that a static drop tests parasitic currents, not only the mean Laplace pressure. κ is sensitive to second-derivative errors and to interpolation across refinement transitions; filtering φ smooths κ but shifts the interface and erases thin sheets.

The explicit bound (Brackbill) is evaluated per level on active interface cells in the GPU reduction:

$$\Delta t \le \sqrt{(\rho_a+\rho_b)\Delta x^3/(4\pi\sigma)}. \tag{29}$$

For water and air ($\sigma = 0.0728$ N/m) it is about 0.22 s at $\Delta x = 3.5$ cm, harmless for the reference wave, but about 1.0 ms at 1 mm, so it controls refined droplets. It scales as $\Delta x^{3/2}$. A semi-implicit version (Zheng 2015, Hyde 2020 [10]) is planned for M4; it changes the operator and its geometric linearisation, not only the time-step check.

The effect test is the $l = 2$ mode of a 3D drop of radius R in an outer fluid (Lamb):

$$\omega_l^2 = l(l-1)(l+1)(l+2)\sigma/(R^3[(l+1)\rho_{\text{in}} + l\rho_{\text{out}}]), \tag{30}$$

which reduces to $l(l-1)(l+2)\sigma/(\rho R^3)$ for negligible outer density, required within ±5 % at stated resolution, damping and amplitude; plus the static jump $2\sigma/R$ and parasitic currents that decrease with Δx. Contact lines need a wall wetting condition; friction alone does not set an equilibrium angle. Variable σ (Marangoni) is out of v1 scope.

9 Particles: narrow band, split, merge, and remap

9.1 Narrow band with an Eulerian interior

The narrow band (Ferstl et al. [17]) is mandatory to meet the memory budget. Liquid particles exist only in a band of thickness $e_b \ge$ support + maximum CFL displacement between reseedings + $2\Delta x$, i.e. $6\Delta x$ by default. **The interior is Eulerian**: `phi_grid` and u_f are advected semi-Lagrangian (BFECC as a builder option) in particle-free cells. On a band face with $m_a < m_{\min}$ but liquid according to `phi_grid`, use $u = u_{\text{Eulerian}}$ and $v_{\text{old}} = u$. A particle touching such a face explicitly sets $\delta = 0$ (counted local PIC); equality of the old and new grid fields alone does not cancel its particle residual. In a $2\Delta x$ overlap, $w = \text{clamp}\left(\frac{-\varphi_{\text{grid}}-e_b+2\Delta x}{2\Delta x}, 0, 1\right)$ blends $u^0 = (1-w)u^0_{\text{part}} + wu_{\text{Eulerian}}$ before the old-grid snapshot. The liquid sign is $\varphi_{\text{liq}} = \min(\varphi_{\text{particles}}, \varphi_{\text{grid}})$: **grid occupancy is authoritative for the interior**, and closed air cavities are kept. Skinning only the particle band would otherwise create a hollow shell with a false inner surface (`surface_hollow_shell`, `deep_tank_no_inner_surface`). When the surface descends, reseed only fully liquid cells from current occupied volume V, current density ρ and advected grid pressure: $J = \text{EOS}^{-1}(p)$, $V_{0p} = \frac{V}{n_{\text{ppc}}J}$ and $m_p = \rho\frac{V}{n_{\text{ppc}}}$. Hencky uses $J = \exp(-\frac{p}{K})$; the selected Tait EOS uses its own inverse. In incompressible mode $J = 1$. Initialise v by gathering and C by the APIC fit on the arrival support, separately from true L. Transfer constituent current volumes, masses and enthalpy using the same ownership rule, logging the Eulerian–particle exchange. Particles deeper than $e_b + \Delta x$ are deleted. Split and merge happen only inside the band. Until M3, closed unsimulated air cavities receive only the capillary boundary pressure and may collapse; the M4 absolute-pressure cavity model below prevents this for resolved cavities.

9.2 Physical inventory and component lineage

The band is a transport/quadrature choice, not the full liquid mass ledger. Simulate owns persistent occupancy, source and boundary inventories, and component ancestry. A semi-Lagrangian sign field may drift geometrically even when it is visually smooth; compare that drift with the physical inventory. Count every region once through the overlap, and derive render volume targets from the owned state and source/flux ledger rather than $\sum J_p V_{0p}$ over the band. At finite sound speed, target volume follows mass/EOS compression rather than an incompressible constant-volume correction.

On a component split, partition the parent target by declared overlap/occupancy weights summing to one; on merge, add targets. Publish uncertainty, the chosen absolute small-component volume floor and cumulative lost volume/count. Apply the same ancestry policy to resolved cavity state. Deep-tank, translating-interface, descending-surface and true-cavity handoff tests precede production acceptance of the narrow band.

9.3 Split 1 → 8

When a particle enters a finer level, with $\Sigma_\delta = D_{\text{parent}} - D_{\text{child}}$ required positive semidefinite (otherwise no split this step),

$$\Sigma_\delta = Q\Lambda Q^T, \quad \delta_c = Q\left(s_1\sqrt{\lambda_1}, s_2\sqrt{\lambda_2}, s_3\sqrt{\lambda_3}\right), \quad s \in \{-1,+1\}^3, \tag{31}$$

$$x_c = x_p + \delta_c, \quad m_c = m_p/8, \quad V_{0c} = V_{0p}/8, \quad v_c = v_p + C_p\delta_c, \quad C_c = C_p, \tag{32}$$

and F, J, history, concentrations, body and CPIC side are copied. Because $\sum_c \delta_c = 0$ and $1/8\sum_c \delta_c\delta_c^T = \Sigma_\delta$, this conserves mass, reference/current volume and linear momentum. Angular momentum and affine-energy claims depend on the row-specific remaining moment: zero residual gives exact energy preservation, while a residual confined to row a's axis preserves the antisymmetric angular moment but removes the corresponding energy. For a uniform 2:1 quadratic split $\delta = \left(\sqrt{3}/4\right)\Delta x_{\text{coarse}}$ per axis. Split is restricted to uniform unmasked parent and child supports and refuses `kAffineOff`, THB transitions, obstacle or cut bands. On MAC the moments are diagonal, so take the componentwise minimum $\Sigma_\delta = \min_{a(D_{pa}-D_{ca})} = \left(\frac{3}{16}\right)\Delta x^2_{\text{coarse}} I$. For the cubic-normal/quadratic-transverse kernel, row a's residual is confined to coordinate a: angular momentum stays exact and the exact energy loss $m\Delta x^2_{\text{coarse}}\sum_a \frac{C^2_{aa}}{32}$ is logged. Also refuse a child in another phase, CPIC side, obstacle or outside the fine level; the parent remains a valid quadrature point.

9.4 Merge n → 1

Only Liquid and Gas particles of the same phase, substance and body, without `kAffineOff`, on uniform unmasked supports before and after, away from THB transitions, obstacle and cut bands, merge ($n \le 8$); Elastic, Granular and Snow **never** merge, while their grid may still coarsen. With $M = \sum m_c$, $\overline{x}$, $\overline{v}$ the mass-weighted means and $r_c = x_c - \overline{x}$:

$$K = \sum_c m_c(r_c r_c^T + D_c), \quad H = \sum_c m_c((v_c - \overline{v})r_c^T + C_c D_c), \quad C^* = HD^{*-1}/M, \tag{33}$$

with D^* the moment at $\overline{x}$ on the destination support, $V_0 = \sum V_{0c}$, $J^* = \sum J_c V_{0c}/V_0$ and $\text{conc}^* = \sum_c \text{conc}_c J_c \frac{V_{0c}}{\sum_c} J_c V_{0c}$, preserving each constituent current volume. This conserves m, V_0, $\sum JV_0$, momentum and angular momentum. The merge is **accepted only if $MD^* - K$ is positive semidefinite**: then $(MD^*)^{-1} \preceq K^{-1}$ and the affine kinetic energy can only decrease, by a Cauchy–Schwarz argument on the stacked moments. The loss is counted (`merge_energy_loss`); otherwise the merge is refused. On MAC the construction is done **component by component**: K_a, row a of H, D^*_a and c^*_a, with diagonal $D_{c,a}$, and the semidefinite test is applied for each a. These formulas derive from the APIC moments of Jiang et al. [1]. Tests: `split_conservation`, `merge_energy_bound`, `merge_counterstream`. Constrained NNLS/MPCR coarsening is deferred.

9.5 Level remap of Eulerian state

On a level change, the Eulerian state (band interior, `phi_grid`, gas scalars, species) is remapped conservatively:

$$V_C = \sum_j V_j, \quad M_C = \sum_j M_j \ \text{(each extensive scalar)}, \quad Q_F = \sum_{f\subset F} a_f u_f, \quad u_F = Q_F/a_F, \tag{34}$$

with prolongation by parent value plus a minmod-limited slope such that children sum to the parent. A face with $a_F = 0$ is blocked. Temperature is recomputed from enthalpy, never volume-averaged; pressure

is a warm start and, at finite sound speed, physical advected state remapped by current-volume averaging. Physical parameters (ρ, μ, σ, Lamé moduli, friction) never change with level; history-dependent moduli are re-evaluated from the remapped state, not averaged. A coarse sign or averaged φ is not a substitute for fractions: coarsening a cell with two interface components could delete a sheet or cavity, which the vetoes of § 2.3 prevent. Faces, components, gauges and multigrid maps are rebuilt on the new generation. Adaptive promotion requires dimensionless partition and first-moment defects at most 10^{-6} in FP32, constant-pressure integrated relative residual at most 10^{-6} with a stated absolute floor, and the pressure solver's volume-scaled divergence limit. Nondimensionalise moments by cell width and test translations, rotations, negative coordinates, both kernels, transition ownership and masked fallback separately. Retained particles keep their affine fit C as an approximation. For $\Delta D = D_{\text{new}} - D_{\text{old}}$, $\log \Delta E = \left(\frac{m}{2}\right) \sum_a c_a \Delta D_a c_a^T$ and the angular change from the corresponding row-dependent antisymmetric moments. Coarsening is vetoed if the positive aggregate energy increase exceeds $\varepsilon_E E_{\text{block}}$ (initial design value 1 %, to measure). Flagged particles have zero affine contribution. Angular momentum across retained-support remap is measured rather than asserted exact.

Thin sheets (decided: below the finest cell they are a meshing matter, not a simulation one). Thickness measured between opposite interfaces enters $h_{\text{interface}}$; thinning regions are reseeded (Ando 2012). Below $\Delta x_{\min}$ the sheet is kept visually by the tracked mesh, without projection, following extrapolated grid velocity.

10 Gas, fire, and explosions

10.1 Reduced low-Mach model

The selected **reduced physical low-Mach** model uses few artist parameters: fixed-composition air ($\gamma = 1.4$), p_0 uniform per pressure component, combustion releasing heat, oxygen not represented. Velocity is carried by APIC particles on the MAC grid; Impulse PIC [11] is a declared transfer variant requiring an ADR, because it transports impulse with its own stretching term. Particles are seeded only in active blocks. The projection is the liquid one without free surface. Gas receives **no full gravity in** ⑤: pressure is measured relative to the ambient hydrostatic state, and buoyancy is the only vertical forcing. Grid state is soot and fuel densities c_{soot}, c_{fuel} (kg/m³), temperature T, and p_0 per component (the reduced thermodynamic model):

$$\partial_t c + \nabla \cdot (cu) = R_c, \quad \rho c_p DT/Dt = \dot{q} + \dot{p}_0 + \nabla \cdot (k \nabla T), \quad \rho_{\text{gas}} = p_0/(R_s T), \tag{35}$$

with $R_s = 287$ J/(kg·K) the specific gas constant, evaluated before ⑧. Fuel, soot and enthalpy use an independently checked conservative flux discretisation with positivity treatment. A semi-Lagrangian or BFECC reconstruction by itself is not an exact mass/enthalpy balance. Recover temperature from the conserved thermodynamic state; limiter smoothness does not replace inventory checks. Reaction: $\omega = \min(k_r c_{\text{fuel}}, c_{\text{fuel}}/\Delta t)$ if $T > T_{\text{ign}}$, $\dot{q} = H_c \omega$, $R_{\text{fuel}} = -\omega$, $R_{\text{soot}} = +y_s \omega$. The divergence source and the closed-domain pressure law are

$$s = ((\gamma - 1)(\dot{q} + \nabla \cdot (k \nabla T)))/(\gamma p_0) - \dot{p}_0/(\gamma p_0), \tag{36}$$

$$\dot{p}_0 = \left((\gamma - 1) \left(\int \dot{q} \, \mathrm{d}V + \oint k \nabla T \cdot n \, \mathrm{d}A \right) - \gamma p_0 \, \mathrm{d}V_{\text{fluid}}/\,\mathrm{d}t \right) / V_{\text{fluid}} \quad (0 \text{ if open}), \tag{37}$$

where the wall conduction term vanishes for the default adiabatic walls. The source includes conduction. $\dot{q}$ (W/m³) comes from reaction rates and sources, **never** from an Eulerian difference of T: a translated non-uniform temperature field must not create spurious expansion (`gas_translation_no_spurious_div`). In a closed box $\dot{p}_0$ restores compatibility, and `sealed_gas_heating` checks $\mathrm{d}p_0/\,\mathrm{d}t = (\gamma - 1)Q/V$ within 1 %. Buoyancy is

$$b = (-a_s c_{\text{soot}} + b_T (T - T_{\text{amb}})) \hat{e}_{\text{up}}, \quad b_T = |g|/T_{\text{amb}}, \quad a_s = |g|/\rho_{\text{amb}} \quad (\text{defaults}), \tag{38}$$

so hot gas rises and soot sinks (`buoyancy_sign`). An explosion deposits energy E as $\dot{q} = E/(V_{\text{dep}} t_{\text{dep}})$; in an open domain without shock the added volume is $\Delta V = (\gamma - 1)E/(\gamma p_0)$, checked within ±25 % (`explosion_added_volume`).

Gas particles are transport markers. Their positive deposition weights are numerical quadrature with explicit initialisation/remapping, not an additional air/fuel/soot inventory. Grid thermodynamics and EOS supply physical density and pressure/viscous inertia. F2 can conserve marker deposition weight but does not thereby prove physical gas mass conservation; scalar flux and enthalpy budgets remain independent.

10.2 Supersonic extension (reserved, not implemented)

Shocks need conservative mass, momentum and energy with shock-stable fluxes; a divergence source is not enough. The retained path is the semi-implicit all-Mach scheme of Kwatra et al. [26], which reuses the multigrid: explicit conservative, limited advection of $(\rho, \rho u, E)$ with the same advector as smoke, then

$$p/(\rho c^2 \Delta t^2) - \nabla \cdot ((1/\rho)\nabla p) = p^\star/(\rho c^2 \Delta t^2) - (\nabla \cdot u^\star)/\Delta t, \quad c^2 = \gamma p/\rho, \tag{39}$$

assembled through the reserved Helmholtz pair of § 6.1 ($V_i p_i/(\rho_i c_i^2 \Delta t)$ on the left, $V_i p_i^\star/(\rho_i c_i^2 \Delta t)$ on the right), and finally correction of u, ρu, E with the same gradient. The extension requires that Helmholtz capability, lazily allocated channels $(\rho, \rho u, E)$ in the compressible branch, an equation-of-state interface $p = (\gamma - 1)\rho e$, the limited flux-form advector, a pluggable time-step policy ($|u|$ or $|u| + c$), and named oracles (Sod shock tube with plateau within 10 %, Sedov blast $R(t) \propto \left(Et^2/\rho\right)^{1/5}$, and bitwise equality with the incompressible projection when $c \to \infty$). Shock–fracture coupling in the manner of Cao et al. 2022 remains a later direction.

10.3 Weak compressibility of liquids (option, M4)

The substance parameter c_{ref} defines the Hencky bulk modulus $K = \rho_0 c_{\text{ref}}^2$. In the reserved pressure compliance use that same tangent modulus: $\rho c_{\text{eff}}^2 = K$, so $c_{\text{eff}}^2 = \frac{K}{\rho}$. Tait supplies its state-dependent $K_{\text{eff}} = \rho \partial \frac{p}{\partial} \rho$ and matching volume-ratio inverse. Reference and current density must not silently define different compliance coefficients.

A per-substance option (M4) makes a liquid slightly compressible through the Helmholtz term. With $K = \rho_0 c_{\text{ref}}^2$, the pressure history is an advected grid field $\tilde{p}^n$, over both band and interior; a newly liquid cell starts at zero. Set $p^\star = \tilde{p}^n$ and solve

$$\left(\frac{V_i}{K_i \Delta t}\right) p_i + \sum_f k_{f(p_i - p_j)} = \left(\frac{V_i}{K_i \Delta t}\right) p_i^\star - \sum_f a_f u_f^\star n_{if} - \text{wall flux} + \ldots. \tag{40}$$

The accepted pressure becomes the next physical history and warm start. Particle J evolves by $J \leftarrow J \exp(\Delta t \operatorname{tr} L)$ and supports density and EOS-consistency diagnostics; recounting particles never defines pressure. The EOS choice is Hencky (default) or Tait: Hencky has $p = -K_0 \ln J$, $J_{\text{EOS}(p)} = \exp\left(-\frac{p}{K_0}\right)$, $K_0 = \rho_0 c_{\text{ref}}^2$; Tait has $p = B(J^{-n_T} - 1)$, $B = \frac{K_0}{n_T}$, $J_{\text{EOS}(p)} = (1 + \frac{p}{B})^{-\frac{1}{n_T}}$, $n_T = 7$ and $K_{\text{eff}} = n_{T(p+B)}$. Enforce $p + B > 0$, including the vapour threshold, and freeze the tangent at the predictor for one declared Picard iteration with measured linearisation error. Reseeding uses the selected EOS inverse and current occupied volume as defined in the narrow-band section.

At $c = \infty$, the compliance vanishes and the incompressible density controller is unchanged. At finite c, use $s_{\text{corr}} = \text{clamp}\left(-\frac{\ln \overline{J} + \frac{\tilde{p}^n}{K}}{t_{\text{corr}}}, \pm s_{\max}\right)$ for Hencky, or generally $s_{\text{corr}} = \text{clamp}\left(-\frac{\ln \overline{J} - \ln J_{\text{EOS}(p^\star)}}{t_{\text{corr}}}, \pm s_{\max}\right)$, zero in the particle-free interior. This controls drift from the advected physical pressure rather than physical compression. Validate $\nabla \cdot u - s = -\frac{p - p^\star}{K \Delta t}$, with the same residual convention as the pressure solve. Nonzero compliance retains physical mean pressure; it requires no Neumann gauge or compatibility subtraction. Positive compliance preserves SPD; any change in PCG iterations is measured, not guaranteed.

Backward Euler damps a resolved acoustic mode by $\frac{1}{\sqrt{1 + (\omega \Delta t)^2}}$ per step in the linear model. A midpoint option applies Crank–Nicolson to the coupled pressure/velocity system; it conserves linearized acoustic energy for fixed geometry, adjoint spatial operators and homogeneous closed boundaries, not arbitrary nonlinear Tait flow or moving geometry. Implicit acoustic stability does not guarantee resolution: use $c \Delta \frac{t}{\Delta} x \lesssim 2$ and at least ten cells per wavelength for visible waves. The default guard $c \geq \max(10 u_{\max}, \sqrt{100 g H})$ targets volume changes below 1 %. Smaller c deliberately makes compression visible and requires checking pressure fronts, water hammer, standing-wave periods and hydrostatic compression.

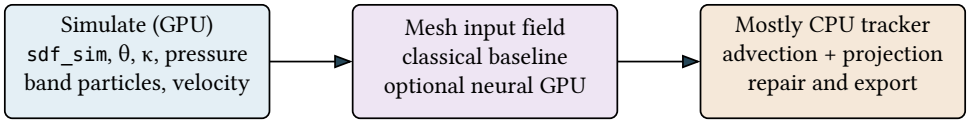
Compliance is assigned only to the simulated fluid fraction; an unsimulated air cavity is counted once, by its boundary pressure. For resolved closed cavities ($R_{\text{eq}} \geq 2\Delta x$), M4 stores absolute pressure $p_{b,\text{abs}} = p_{b,\text{abs},0} \left(\frac{V_0}{V}\right)^\gamma$, and supplies gauge boundary pressure $p_\Gamma = p_{b,\text{abs}} - p_{\text{atm}} + \sigma\kappa$. At closure initialise to p_{atm}; an initially submerged spherical cavity starts at $p_{\text{atm}} + \rho |g| d + 2\frac{\sigma}{R_{\text{eq}}}$. The explicit cavity restriction is $\Delta t \leq \frac{0.5}{\omega_{\text{cav}}}$, $\omega_{\text{cav}} = \frac{\sqrt{3\gamma \frac{p_{b,\text{abs}}}{\rho}}}{R_{\text{eq}}}$. Maintain cavity volume and identity lineage across splits and merges; sub-resolution disappearance is logged. Cavitation uses an active-set Dirichlet vapour-pressure clamp, with an explicit gauge convention. Acoustic energy $\sum V \frac{p^2}{2K}$, density and EOS-relative drift are logged. Realistic sound remains an acoustic post-process. This MAC/APIC/free-surface synthesis and all acoustic targets require implementation and independent validation.

10.4 Flow-map transport for gas (under a decision gate)

A flow map tracks the map from initial to current position and its Jacobian, and transports the **impulse** $m_c = T_{ac}^T m_a$ with $T = F^{-1}$ over n steps without re-interpolation, which respects Kelvin circulation and preserves vorticity much better [29]. Its F is exactly the MPM F and evolves the same way, but there is no constitutive law or force from it. **IPIC is PFM with $n = 1$.** This is a transfer variant, not a new solver: the MAC grid, projection, fixed-point P2G and transactions are unchanged, and a pluggable `Transport` stage offers `APIC` (default), `Impulse(1,1)` (IPIC) and `Impulse(n_L>1, n_S)` (PFM). In G2P, $T_{bc} \leftarrow T_{bc}\big(I - \Delta t L_p\big)$ with the same ∇N, $m_c = T_{ac}^T m_a$, $\nabla m_c = T_{bc}^T \nabla m_b T_{bc}$, and path-integrated forces $\Lambda \mathrel{+}= \Delta t T_{ac}^T \big(f_{\text{buoy}} - 1/2 \nabla\ |u|^2\big)$; P2G deposits $N' m_p \Big((m_c - \Lambda)_a + \nabla m_c \cdot r' \Big)$. Maps reinitialise every n_L (long) and n_S (short) steps, and are forced to $n = 1$ locally on merge, reseeding, level change or THB junction, within $2\Delta x$ of bodies, cuts or contact, when $\text{cond}(T) > \kappa_T$, or where `kAffineOff` is set. Splits copy T, m_a, ∇m and Λ. In the Impulse branch $\alpha = \beta = 0$; n_L, n_S are internal; no grid velocity-history buffer is kept.

Liquid stays APIC/ASFLIP: the cited sources do not validate flow maps combined with a two-phase ghost-fluid free surface, surface tension and a narrow band. Compressible flow keeps the flux-form advector. The low-Mach delivery baseline is APIC. Optional IPIC research must beat that baseline measurably (vortex-ring energy at 2 s at least 5 points above APIC), otherwise the branch is removed. A later PFM experiment uses its own adoption gate: at least five leapfrog passes against at most three, ring energy ≥ 90 % at 2 s, cost ≤ 1.5 × IPIC and at most +100 allocated B per gas slot (97 B raw payload; the separately aligned map-age array costs 4 B per slot rather than one); **determinism**, `determinism_bytes` passes with reinitialisation, with CPU/CUDA parity on integral metrics only. A transport interface may preserve this extension path, but no unused flow-map arrays are allocated in the APIC baseline. Nothing here is implemented or measured.

11 Surface: tracked mesh and multiresolution



Data flow is one-way: Simulate never reads `mesh/`; rerunning Mesh leaves `sim/` unchanged.

Figure 6: Stage responsibilities. Simulate owns its interface geometry. Mesh is a pure consumer; Multires is an optional consumer of Mesh.

11.1 Responsibilities and inputs

Simulate owns `sdf_sim`, θ and κ and never reads Mesh. Mesh is a pure consumer: rerunning it with other settings changes no byte of `sim/` (`mesh_rerun_sim_hash`). Mesh runs on the **CPU**, one frame behind Simulate and in parallel with it; a GPU port (Chentanez et al. 2016) is triggered only if the budget gate of § 14.4 fails. Simulate exports per frame: `sdf_sim` (band at Δx, interior sign from authoritative Eulerian occupancy), band particles (positions, radii), velocity extrapolated three cells into air with a validity mask, obstacle SDF and velocity, and per-component labels and liquid volumes (FP64).

11.2 Motion across an output interval

Mesh and Whitewater consume a declared `MotionInterval`, not merely the velocity at the final frame. Start with timestamped carrier samples and a declared interpolation on fixed caches. Bound transport error on translated and folding surfaces; increase sampling only when that benchmark requires it, or compare a compact transport map if storage dominates. Exporting every substep without a measured storage budget is not the default. Reconstructing a tracked arbitrary frame requires a prior MeshState/checkpoint and deterministic replay.

11.3 Target field and tracked mesh

Candidate geometric target `sdf_render` (OpenVDB on CPU, at Δx): a Zhu–Bridson field on band particles, united by min with `sdf_sim` below the band, so there is no false bottom surface and closed air cavities survive. Compare isotropic and anisotropic kernels and constrained skinning before selecting production smoothing. The current candidate variant of Bhattacharya et al. [13] applies K Laplacian iterations in the band with φ bounded between φ_{int} and φ_{ext} from particle radii, then redistancing. There is **no temporal filter** on the SDF (it caused ghosts and fps dependence); temporal coherence comes from the mesh.

Tracking. Vertices advect by RK2 using `MotionInterval`, with spatial sub-steps of at most $0.5\Delta x$ and separately checked temporal error. Bounded projection

$$x \leftarrow x - \lambda\varphi\nabla\varphi/|\nabla\varphi|^2, \quad \lambda = 1 - \exp(-\Delta t/\tau), \tag{41}$$

with τ in seconds (so independent of fps) and displacement capped at $\min\big(0.5\Delta x, 0.5\Delta x\Delta\frac{t}{\tau}\big)$ (a physical correction-rate cap), is refused if $|\nabla\varphi| < 0.5$, if it crosses another component, or if the vertex lies in an **unresolved sheet**: a ray cast along $-n$ finds the mesh again within $2\Delta x_{\text{sdf}}$. Bound root search or trust-region projection by feature separation and phase identity, with obstacle exclusion, so opposite sheet sides cannot converge onto one root. Such vertices only advect; `thickness` is measured by that ray, and the sheet ruptures only below t_{rupture}, the hole being a local repair. Tangential relaxation then local split, collapse and flip target an edge length of Δx, finer where curvature demands; no operation creates a non-manifold edge, and edge quality is restored by remeshing; physical surface stretching under advection is allowed.

Topology changes follow Wojtan et al. [27, 28]: the mesh's own signed occupancy is sampled on a Δx grid; a cell is **complex** if one of its edges is crossed more than once or the mesh self-intersects there. Repair happens only on a **real intersection** (merge) or a rupture decided above: the complex cells, dilated by one and closed to compatible boundaries, are cleared, rebuilt by marching cubes on exact triangle distance on a local grid $\Delta x_{\text{repair}} = \Delta\frac{x}{2^k}$, and stitched. Choose the smallest $k \le 4$ with $2\Delta x_{\text{repair}} \le t_{\min}$ in the region. If no such grid resolves the sheet, rupture locally with W-MESH-SHEET-LOSS and log lost area. The output invariant is a closed, oriented, manifold, self-intersection-free surface; after three failures the component is re-extracted from `sdf_render` with W-MESH-REPAIR-FALLBACK and logged lost sheet area. Every repair is logged (region, dead and born IDs).

Attributes. Survivors keep `id`; new IDs come from a counter in sorted operation order (deterministic). `age` is zero for new vertices, `wetness` is interpolated, `thickness` recomputed; `velocity` is geometric displacement over Δt (motion blur) and `fluid_velocity` is sampled from the field. Stable IDs measure identity, not quality: FB3 asks for ≥ 90 % surviving IDs outside logged repair regions for a drop into a pool, ≥ 95 % in calm phases.

11.4 Volume lock, droplets, preview

Per connected component, correct the delivered vertices rather than the target field:

$$\delta = \operatorname{clamp}\left(\frac{V_{\text{sim}} - V_{\text{mesh}}}{A_{\text{mobile}}}, \pm 0.25\Delta x\right), \quad x_v \leftarrow x_v + \delta n_v. \tag{42}$$

A_{mobile} includes only vertices outside obstacle contact and unresolved sheets. Remeasure volume on the final mesh and check global self-intersection after each correction; halve δ if it introduces an intersection. Allow at most three iterations, then W-MESH-VOLUME-INFEASIBLE; IDs remain unchanged. Raw simulation and corrected mesh volumes are published separately, with no compensation between components. A component smaller than $2\Delta x$ becomes a liquid droplet (`droplets`, distinct from spray) counted in volume, returning to mesh above $3\Delta x$; it has exactly one representation. Preview extraction uses `sdf_render` without tracking.

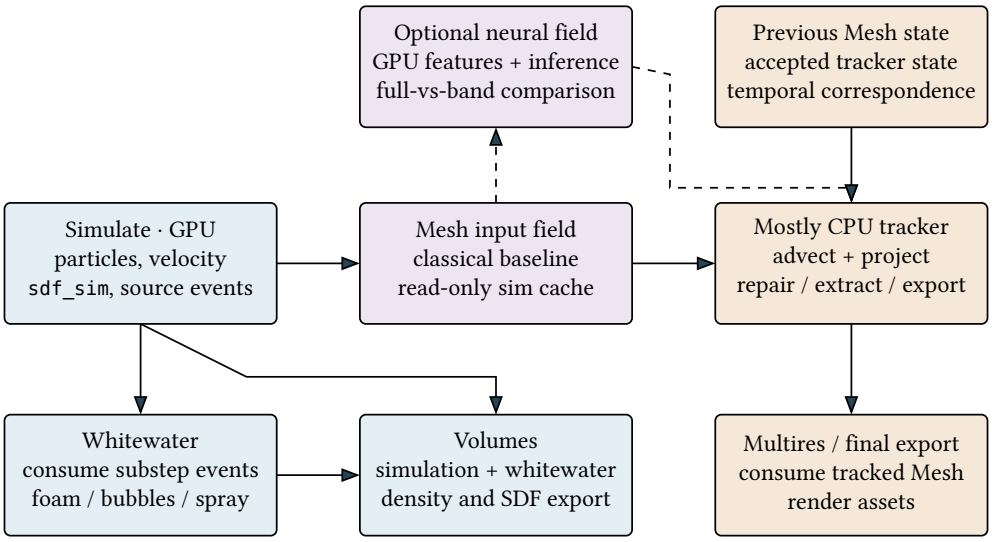
11.5 Shared phase interfaces

Immiscible liquids require one labeled shared phase-pair interface. Independent reconstructions of each phase must not create overlaps or gaps. Triple junctions need a tested multi-material method; one plane per simulation cell does not certify them. Introduce alternative tracking or representation only when a measured shared-interface failure justifies it.

11.6 Cage and native Multires (conditional M3 delivery)

When requested by the reference benchmark, validated cage/native Multires delivery belongs to M3. A candidate quad cage is built by manifold dual contouring on `sdf_render` at 4–8 Δx, per component. A **certificate** requires the same number of components and Euler characteristic as the tracked mesh, Hausdorff distance at most $0.5\Delta x$ after projection, no flipped face, and global triangle–triangle intersection checks on the projected cage and finest rendered subdivision. Every intermediate delivered LOD must also pass its own check or a conservative geometric bound; a failing component is delivered from the tracked mesh, and the full tracked mesh is always delivered. Delivery is either a Geometry Nodes group (subdivision then two bounded projection steps using triquadratic grid sampling) or a true Blender Multires: the solver delivers an already valid cage, Blender subdivides, a copy is projected, and `multires_reshape` transfers it. Existing Multires detail requires unchanged receiving topology and correspondence, since Reshape requires

identical topology and indices. A topology event requires explicit detail transfer/rebuild or component fallback. Persist cage generation and interval identity, connectivity, hierarchy/level mapping and detail correspondence, checkpointing state needed for replay. A stateless per-frame cage rebuild does not establish preserved native detail. A pinned headless reader/export spike must demonstrate the intended animated or snapshot delivery without a per-render-frame Python handler before native mode is claimed. Subdivision cannot create a missing handle or droplet; the certificate is what excludes such cases. The accepted trade-off: topology changes only at real intersections; sheets thinner than Δx are kept for rendering, their physics resolved down to $\Delta x_{\min}$ and then carried by the mesh alone (decided).



Simulate never reads Mesh. Whitewater and Volumes do not require the tracked mesh.

Optional neural GPU work shares device memory and compute with Simulate: schedule and measure contention.

Figure 7: The simulation cache has independent consumers. A neural field backend is an experiment inside Mesh; GPU work competes with the simulator and changes the measured pipeline cost.

11.7 A controlled reconstruction experiment

The render reconstruction is a consumer of saved simulation data. Its first experiment compares one geometric reference with the published Zhao–Shinar–Schroeder CNN [14], using identical saved particle frames. Evaluate the CNN twice: once with the complete saved particle set and once with the actual intended reconstruction band. Keep network architecture, weights, pruning rules, scalar-grid spacing and query locations fixed. This separates a missing-input error from the model's own reconstruction error. A favourable result may justify a later cache or backend decision; it does not automatically change the simulation band or select a neural production dependency.

The source-inspected artifact is the 2025 PACMCGIT paper, Article 51, DOI 10.1145/3747856. The inspected PDF SHA-256 is

```
d314b6455cea5b680dcdf78737d5a021e2657e029ce1b5aac8f935e0355a81a2
```

and the published source ZIP SHA-256 is

```
bc4304606c6b378bfd06dec928025e38d8cf6fd767f70a43d8bdc0f56c3e16ca
```

The source archive includes a pretrained model and ten dam-break frames. It was inspected, not built or executed. No third-party archive, source tree or pretrained weights are stored in this repository. The article's CC BY 4.0 statement does not establish software or weight rights: the ZIP lacks a standalone license, references separately obtained PhysBAM, and supplies no identified separate weight license. Distribution or adoption requires resolving those terms.

The shipped network receives an 8^3 scalar patch: four unpadded 2^3 convolutions, with channels $1, 64, 128, 256, 512$, a size-two pool after the second convolution, then a $512, 256, 128, 1$ multilayer perceptron. There are 1,542,081 parameters including biases. The supplied ONNX shape is fixed at $[1024, 1, 8, 8, 8]$ with output $[1024, 1]$; the native CUDA batching and final padding also assume 1024. Reusing the published path therefore requires coordinated batching changes for other batch sizes, rather than assuming arbitrary dynamic input.

11.7.1 Source-derived support of the shipped configuration

Let h_φ denote the CLI output-grid spacing and $h_N = \text{stride}\ h_\varphi$ the spacing of an inference subgrid. In the shipped stride-two evaluation, $h_\varphi = \frac{h_N}{2}$ and the quartic kernel radius is $2.5h_N$. For one evaluated scalar-grid query q, the eight feature positions along each axis are $q + (j - 3.5)h_N$, $j = 0, \ldots, 7$. A feature depends on contributing particles, whose weights divide by an interpolated particle-density estimate. That estimate in turn depends on more particles. Consequently the dependency chain is

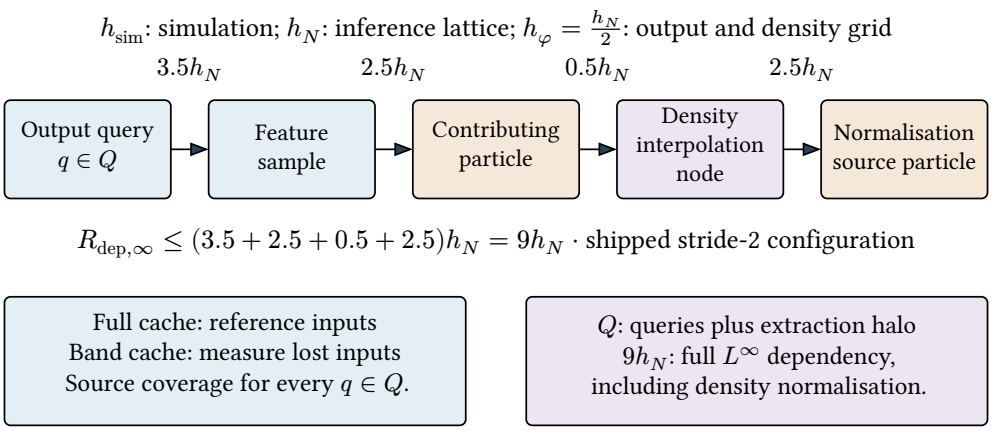
$$\text{query} \to \text{feature sample} \to \text{particle} \to \text{density node} \to \text{density particle}, \tag{43}$$

and a conservative axis-aligned neighbourhood is

$$R_{\text{dep},\infty} \le 3.5h_N + 2.5h_N + h_\varphi + 2.5h_N = 9h_N. \tag{44}$$

Here h_{sim} is independent of h_N and h_φ. This is a source-derived L^∞ enclosure around each evaluated scalar query for that configuration. It is neither a minimum support nor a Euclidean radius; an enclosing Euclidean ball may need $9\sqrt{3}h_N$. The source's accumulation stencil is specialised to stride two, so a different stride, kernel, feature or interpolation requires a new derivation. Query-selection uncertainty, complete extraction cells and postprocessing need additional halos. A particle band measured by surface distance cannot be identified directly with this query-centred box.

Keeping only particles that directly touch the feature patch omits the density denominator's dependence. Replacing missing bulk by caps, coarse particles or hand-filled density changes the input distribution until a full-cache comparison establishes equivalence. The published occupied-cell pruning and skipped-node values must also match across the full/band runs; they are not a proof that bulk data are irrelevant. Measure the actual retained support, field error and surface error before proposing any cache enlargement.



Negative interior output fill cannot restore lost CNN inputs or density denominators.

A surface-distance particle band needs its own coverage check; this bound does not set h_{sim} or widen that band.

Figure 8: The source-derived CNN dependency follows feature construction and its density denominator. The $9h_N$ bound is a query-centred axis-aligned enclosure for the shipped stride-two configuration, distinct from a surface-distance particle band.

Before promotion, compare tiled and monolithic inference with complete dependency halos, a fixed global origin, matched pruning and identical batching/order. Filling known deep liquid negative in the output field can prevent an extraction shell but does not repair missing network inputs. If any coverage or fidelity gate fails, retain the geometric backend; band widening, new occupancy channels or retraining are distinct experiments whose total simulation/cache cost must be measured.

11.7.2 Cost, quality, and a bounded hybrid backend

On its reference platform, the paper reports 2.21 s for one two-million-particle frame on an RTX 4090 and a 32-core Threadripper PRO 5975WX with 512 GB RAM: 1.01 s model inference, 0.90 s input/output and model loading, and about 0.30 s CPU work. The 1.31 s figure excludes loading and I/O. The executable produces scalar arrays or VDB; its README describes later polygon conversion in Houdini. These timings therefore do not certify an end-to-end delivered polygon mesh, and they are not FluidiTy measurements. Record feature construction, inference, extraction, repairs, transfer, I/O, cold start, steady state and peak memory separately on the target machine.

Test translated smooth surfaces, sheets and gaps of thickness $2h, h, \frac{h}{2}, \frac{h}{4}$, droplets, wall contact, cavities, folding sheets and topology changes, including adaptive particle density. Report scalar and zero-set disagreement with full-data CNN separately from geometric-reference quality: distance and normal error, raw per-component volume bias, lost components, cavity/thickness retention, temporal artifacts and residual after known advection. Evaluate manifoldness and global intersections after extraction. Neither a smooth field nor a stable-looking sequence establishes tracked vertex identity.

A later hybrid research backend may use a learned target field inside the existing tracked-mesh pipeline while retaining explicit advection, bounded projection, local repair, IDs and volume correction. This changes field construction only; it does not turn a network into a collision detector, a topology proof, a physical capillary model or a substitute for missing simulation history. Compare it with the full classical pipeline at matched resolution and memory, with a deletion path if the measured benefit does not justify complexity. No automatic neural adoption follows from implementing the experiment. If promoted, persistently load one versioned model and freeze its execution artifact for the bake. Fingerprint model bytes, preprocessing, physical scales, query origin/order, batch policy, precision and inference engine; a model name alone does not fix engine tactics. Checkpoint the prior mesh, backend identity, lineage and ID counter. CPU model parity uses the same model and preprocessing; geometric fallback is a separately named backend. Bound immutable frame queues, copy/event lifetimes, backpressure and the shared CPU thread pool. Enable concurrent neural and simulator kernels only after measuring benefit on their shared GPU.

Candidate	Possible role	Evidence boundary
Anisotropic kernels [30]	Geometric particle-to-field comparator.	Must retain opposing sheets, cavities and occupied interior.
Zhao CNN [14]	Liquid-particle scalar reconstruction; first neural comparator.	Full versus actual-band inputs, license status and end-to-end cost remain gates.
Neural PLS [35]	Oriented interface particles, training and hybrid tracking representation.	Changes the representation; inspected author page establishes no available code/license.
Hermite-NGP [36]	Derivative-aware hash-grid representation.	PDE/SDF evidence does not establish a trained dynamic particle-surface backend.
Subgrid marching tetrahedra [37]	Extraction from reliable multiple edge intersections.	Cannot invent trustworthy particle geometry or missing intersections.

Table 2: Field inference, dynamic tracking and extraction are separate experimental choices.

12 Whitewater

A grid variant of Ihmsen et al. [12] uses specific potentials independent of particle mass. With $\Phi(I, \tau_{\min}, \tau_{\max}) = \mathrm{clip}\left(\frac{I-\tau_{\min}}{\tau_{\max}-\tau_{\min}}, 0, 1\right)$ and ordered unit-compatible thresholds,

$$I_k = |u_c\frac{|^2}{2}, \quad I_{\mathrm{ta}} = \sum_{j\neq c}\big(|u_c - u_j| - (u_c - u_j)\cdot\hat{x}_{cj}\big)W\big(|x_c - x_j|\big), \tag{45}$$

$$I_{\mathrm{wc}} = \max(0, \kappa_c)[u_c\cdot n_c \geq 0.6|u_c| \ \text{ and } \ |u_c| > 0], \quad \Phi_c = \Phi(I_k)(k_{\mathrm{ta}}\Phi(I_{\mathrm{ta}}) + k_{\mathrm{wc}}\Phi(I_{\mathrm{wc}})). \tag{46}$$

Use the 26 neighbouring source cells and $W(r) = \max\left(0, 1 - \frac{r}{2r_{\mathrm{ww}}}\right)$. Exclude self and zero-distance pairs. The potential units are $\mathrm{m}^2/\mathrm{s}^2$, m/s and m^{-1} respectively; the k coefficients are dimensionless. Invalid normal or curvature raises an explicit diagnostic, never a NaN hidden by multiplication by zero. Normals and curvature come from `sdf_sim`; zero velocity gives no crest source without normalising it. The source grid has fixed scene spacing r_{ww} in metres, independent of quality, camera and AMR; each region has one owner. A source requires valid velocity and occupied liquid volume $V_c > 0$.

Step ⑫ exports one trial event $(t_s, x_c, u_c, n_c, V_c\Phi_c\Delta t_s)$ per nonzero source cell and accepted sub-step, without temporal averaging. Two opposite bursts within a frame remain separate. Capacity is checked (E-WW-SOURCE-CAPACITY); rejection discards its events and source state. The event weight stores occupied volume at birth, $w_s = V_{c(t_s)}\Phi_c\Delta t_s$, rather than looking it up at the final frame. Whitewater emits $n = q\frac{w_s}{w_{\mathrm{out}}}$ plus a carried remainder, with q in $\mathrm{m}^{-3}\cdot\mathrm{s}^{-1}$, dimensionless potentials and w_{out} physical particles per output point. Each point is born at the declared quadrature time t_s and integrated through `MotionInterval` to output time. Compare counts and position/velocity/age distributions under output-rate changes; fixed-

frame RNG keys alone do not prove fps independence. Changing the foam amount does not rerun Simulate. Points are distributed in a cylinder of radius r_{ww} with source velocity plus radial noise.

Depth d in `sdf_sim` gives spray for $d > r_c$, bubbles for $d \leftarrow r_c$, foam otherwise, with hysteresis $0.5r_c$. For frozen driving fields, spray and bubbles use the exact velocity and position update

$$v^+ = v_{\text{eq}} + \big(v - v_{\text{eq}}\big)e^{-k_d \Delta t}, \quad x^+ = x + v_{\text{eq}}\Delta t + \big(v - v_{\text{eq}}\big)\frac{1 - e^{-k_d \Delta t}}{k_d}, \tag{47}$$

with $v_{\text{eq}} = v_{\text{air}} + \frac{g}{k_d}$ for spray and $v_{\text{eq}} = u_f - k_b \frac{g}{k_d}$ for bubbles; k_d is s^{-1} and k_b dimensionless. Evaluate with `expm1` and analytic zero-drag limits. Foam follows $x^+ = x + \Delta t u_f$ and expires by age. Outside velocity validity, spray remains ballistic with exact ambient-wind drag, bubbles retain their class and explicitly use $u_f = 0$, and foam requires its exported carrier halo: a missing field raises E-WW-MISSING-CARRIER with coordinate and time, requiring cache repair or re-export, without class change or ballistic fallback. Never interpret an invalid VDB sample as a valid zero velocity. Collisions use the obstacle geometry and continuous-motion contract.

Coupling is one-way: Whitewater reads `sim/`, never `mesh/`; its points do not displace liquid and cannot model aeration feedback. Philox emission keys use (source cell, global accepted sub-step, ordinal), and dynamics use (id, frame). Geometry Nodes projects foam from `whitewater/`; Mesh never reads it. Multi-scale bubbles remain a future direction.

13 State, storage, and memory

13.1 Particle state

Particles are AoSoA in packets of 32, one 128 B-aligned array per physical field, one stream per (block, continuum). **Persistent** fields survive the sub-step and are checkpointed; **scratch** fields are rebuilt every sub-step. A particle index changes when sorting; `id` never does.

Field	Type	Unit	Continua	Lifetime
`x`	3 f32	m (absolute)	all	persistent
`v`	3 f32	m/s	all	persistent (FLIP)
`C`	9 f32	s^{-1}	all	persistent (C, not B)
`m`	f32	kg; numerical marker weight for Gas	all	persistent; split/ merge
`V0`	f32	m^3 (reference)	all	persistent
`id`	u64	—	all	persistent, never reused
`tag`	u32	packed bits	all	persistent: substance u16, phase u8, flags u8
`J`	f32	1	Liquid, Gas	persistent
`F_E`, `log_J_pl`	9 f32, f32	1	Granular, Snow	persistent
`F`	9 f32	1	Elastic	persistent
`body`	u16	body index	Granular, Snow, Elastic	persistent
`affinity`, `side`	2 u32	bits	Elastic (M6); cuttable Granular/Snow	persistent
`conc[k]`	k f16	volume fraction	miscible Liquid	persistent, $\Sigma = 1$ renormalised
`temperature`	f32	K	when thermal branch is active	persistent
`dst`, `cell`	2 u32	index	all	scratch (sort)

Table 3: Persistent particle state (candidate ABI). Common base 80 B; Liquid 84 B, Granular/Snow 122 B, Elastic 126 B, Gas 84 B, plus 8 B scratch. The packed tag is required for the 84 B Liquid stride (2688 B per packet); separate narrow arrays would waste alignment padding. The memory budget keeps 88 B for Liquid as margin; generated ABI accounting includes body padding and optional map-age arrays. Gas in Impulse mode would add optional flow-map slots.

Density is $\rho = m/(JV_0)$ for Liquid and $m/\left(\det F_E J_{\mathrm{pl}} V_0\right)$ for Granular and Snow. Liquid J feeds the density correction only inside the band; the particle-free interior has no particle drift sensor; physical pressure and occupancy history persist there. Surface tension is a property of a **phase pair**. Whitewater points are a different record owned by the Whitewater stage (`kind`, `age`, `id`, `velocity`, `radius`, hysteresis state, RNG key).

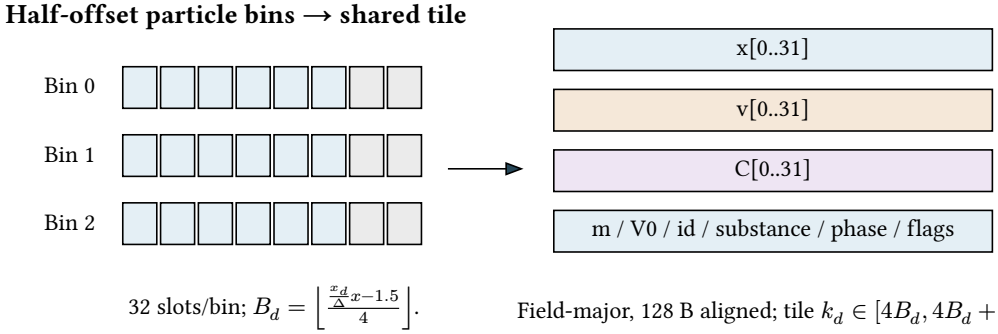


Figure 9: AoSoA organisation. A block/continuum stream owns compact bins; each field is contiguous across lanes. Allocated capacity, not live count, sets memory.

13.2 Grid state

Channel	Location	Type	B/cell	Lifetime
`u`	faces	3 f32	12	persistent (Eulerian band interior), versions n / n+1
`phi_liquid`	cell	f32	4	persistent, versions n / n+1
`p_prev`	cell	f32	4	persistent, trial-versioned (warm start and physical pressure at finite c)
`acc_m`, `acc_q`, `acc_f`	faces	3 × 3 i64	72	P2G scratch; mass decoded in place, later reused for PCG/geometry
`u_old`	faces	3 f32	12	scratch; live through G2P
trial `u`, `phi`			16	transaction
peak per active cell			**≈ 136**	

Table 4: Grid channels (candidate storage). Sparse tables exist only in their blocks: multi-velocity contact ($K = 4$), CPIC (12 B/face), gas scalars, future compressible channels (ρ, ρu, E) and temperature only when the corresponding branch is active. Schema extensibility does not allocate inactive fields. Multigrid coarse levels add about 3 B/cell.

Block deactivation or reactivation conservatively transfers persistent fields and cannot erase Eulerian occupancy, pressure history or gas inventory by resetting a transfer accumulator. Active sets are nested (particle blocks ⊂ P2G neighbour blocks ⊂ advection blocks), compacted by scan and **sorted by key**. A block key, dense index, level, neighbour indices, active masks and a topology generation form the topology; neighbour maps are built once per generation so that kernels never hash per stencil entry.

13.3 Memory budget of the reference wave

Assumptions (design choice, to be re-measured at M3): 3 m depth, surface area doubled while breaking, the whole surface at the finest level, a $6\Delta x$ band, 8 particles per cell, i.e. about 12.5 M particles and a capacity of 16 M.

Item	Basis	GB
Liquid particles	16 M × (2 × 88 + 8) B × 1.15 (packets)	3.4
Active grid	≤ 7.5 M cells × 136 B	1.02
Topology and hash	≈ 117 k blocks	0.03
Obstacle SDF and velocity	near obstacles	≤ 0.12
CUB temporary storage	queried at allocation	0.2
Total		**≈ 4.8**
With 25 % margin		**≈ 6.0**

Table 5: Reference-wave VRAM estimate (provisional sizing) against the approximate 11 GB job-wide cap, $\min(0.70 \cdot \text{total}, 0.80 \cdot \text{free at job start})$.

Without a narrow band at 3.5 cm (about 90 M particles) the particle term alone is about 19 GB, so the narrow band or adaptivity is mandatory. The uniform-plus-band fallback has a provisional 6.8 GB estimate. `fluidity check` estimates the budget from measured occupancy; Whitewater, Mesh and Volumes run in separate processes. The two particle buffers are the double buffer of the transaction (§ 14.2). Library scratch must be queried at maximum capacity, not guessed. Enforce one stable job-wide ceiling over simultaneous processes, neural weights/activations/features/workspaces, stage buffers and in-flight copies. A generated liveness table must establish arena reuse; resetting scratch does not prove overlapping lifetimes fit. The approximate 6 GB estimate excludes unmeasured neural/event/liveness costs and must be replaced by observed peaks.

14 Execution contracts

14.1 GPU execution

Kernels launch over **capacity** (or grid-stride) with early exit on a device-side live count. Arena pointers pass through a device control block, so captured execution growth between frames first attempts

`cudaGraphExecUpdate` and recaptures/reinstantiates if changed CUB nodes cannot update. CUB primitives are captured at capacity with neutral inactive inputs, queried scratch at capacity and a pinned CCCL version; conditional handles are initialised on every entry. The measured graph-capture optimisation, once validated, nests `while` (sub-steps) ⊃ `while` (attempts) ⊃ `if` (status), with no allocation in conditional bodies [16]. G2P2G fusion (G2P of step n joined with P2G of $n+1$ [5]) is allowed only with the same topology generation, no scheduled emission or adaptation, a continuum without Δt-dependent constraint, and a certified uniform support; otherwise G2P and P2G run separately, and the early deposition belongs to the trial. There is **no allocation inside a sub-step**; capacity differs from live count, and overflow is CAPACITY, never a silent loss. Precision: fixed-point i64 P2G; FP32 state; FP64 for clocks, conservation sums, PCG dot products and the coarse level; BF16 forbidden. Stress (matrix work, SVD), pressure (global sparse solve) and mesh repair (irregular topology) have different work profiles and are profiled as such [15].

14.2 Transactions

A committed state A and a trial B. Particles are double-buffered: the trial writes B, A is never touched, and a rejection costs no copy. Persistent Eulerian fields (`u`, `phi_liquid`, `p_prev`, gas inventories and physical source inventory) have versions n / $n+1$, swapped on acceptance. Whitewater birth events are private trial append records; acceptance appends them to the accepted frame stream and rejection discards them. Artist scaling and fractional whitewater emission remainders belong only to the rerunnable Whitewater stage. Topology carries a generation g and is double-buffered: accepted A remains immutable, trial B owns its keys, maps and fields. After ACCEPTED, adaptation is a separate transaction reading the accepted state and writing the free buffer; local refusal leaves the particle unchanged, capacity invokes the frame restart. A device control block (≈ 1 KB, copied into the trial) holds FP64 time, Δt, next `id`, fractional physical particle/fuel source credits (FP64), RNG counter, attempt count and status. Statuses:

1. ACCEPTED: versions swap; the clock advances exactly once.
2. RETRY_DT (CFL violated a posteriori, pressure not converged, $\det F \le 0$, inadmissible F, trial NaN): restart from A at $\Delta t/2$, at most twice, then an E- error.
3. CAPACITY: leave the graph, grow arenas, restart the frame from the frame-start snapshot (§ 14.4), at most once, then E-MEMORY-BUDGET.
4. FATAL (CUDA error, overflow, corruption): stop.

An incompatible pressure right-hand side is an error, not a retry. Repeatedly halving Δt never repairs a full table.

14.3 Determinism and CPU/CUDA parity

Bitwise reproducibility on the same GPU, driver and build relies on: every mass, pure-momentum and force contribution quantised per particle before any sum, $Q = \operatorname{rint}\left(\frac{c}{s_c}\right)$ in i64 with a logged scale per channel. Before conversion require finite c and check $|\frac{c}{s_c}| \le 2^{62-n_{\text{cap}}}$, where 2^n_{cap} bounds total particle capacity; this bounds absolute sums independently of cancellations. Failure is FATAL E-P2G-RANGE, with particle and channel, never saturation. Actual contributions include affine velocity, mass and stress; a velocity cap alone is insufficient; stable counting sort; active blocks sorted by key at every generation; floating reductions in a fixed tree (CUB `run_to_run`), with floating scans forbidden without that guarantee; IDs allocated by scan on (parent id, ordinal); Philox RNG with one key per consumer, (id, frame, accepted sub-step) for particles and (source cell, global accepted sub-step, ordinal) for whitewater emission; serialisation by increasing `id`. CPU and CUDA share the same `FLUIDITY_HD` kernel bodies and quantisation and are compared by tolerance: per kernel, P2G within one quantum per contribution and G2P $|\Delta x| \le 10^{-6}$ m; per sub-step, $\|\Delta p\|_\infty \le 10$ tol p_{ref} and $\|\Delta v\|_\infty \le 10^{-4} v_{\text{ref}}$; per trajectory, integral metrics only (dam break at 0.5 s: front ±1 Δx, volume ±0.1 %, kinetic energy ±2 %).

14.4 Stages, manifests, and checkpoints

Stages are separate and rerunnable: **Simulate** (GPU) writes `sim/` (band VDB grids `sdf_sim`, velocity and mask, obstacle SDF; band particles; components; sparse whitewater sources; stats, logs, trace; full particles at checkpoints and, as an advanced debug option, every frame); **Mesh** (CPU, frame $N-1$, in parallel) writes `mesh/` (Alembic tracked mesh with `id`, `velocity`, `fluid_velocity`, `age`, `wetness`, `thickness`, `substance`, `phase`, plus `droplets`); **Whitewater** writes `whitewater/` points; **Volumes** writes foam, bubble, liquid SDF, velocity and gas density, temperature, fuel and flame VDBs; **Multires** writes the validated cage and `sdf_render`. Rerunning a stage invalidates only its consumers; rerunning Mesh invalidates only Multires.

Each stage has a manifest (candidate FlatBuffers representation with a JSON mirror) holding status (`baking` with pid and heartbeat, `complete`, `cancelled`, `error`; a `baking` manifest whose process died reads as `crashed`), completed frames, and a content fingerprint with a documented hash algorithm over schema, the stage's own **resolved** settings in canonical serialisation, external data hashes, build and backend hashes, and upstream **content** sums (never manifest bytes). Publication is atomic (temporary file, rename, manifest last); the signal handler only writes a preformatted buffer with `write(2)`. A frame-start snapshot copies persistent state asynchronously to pinned host RAM (≈ 1.5 GB). An event/fence prevents reuse or mutation of source buffers until the copy completes. Its version includes topology, pressure history, control/source state and hysteresis; it serves CAPACITY restarts, crash dumps and checkpoints.

Stage	Checkpoint content	Resume
Simulate	persistent particle fields, `u`, `phi_liquid`, `p_prev`, gas scalars, block keys and levels, hysteresis ages, control block, body and blade tables; build, schema, GPU and driver hashes	from checkpoint $c \leq$ last published frame p; frames $c+1..p$ are recomputed and compared to published sums without republishing; a mismatch gives W-RESUME-NONDETERMINISTIC
Mesh	mesh $N-1$, volume-lock state, ID counter, repair log	from the last published frame
Whitewater	live points (class, hysteresis, age, key), RNG counters, emission remainders	from the last published frame
Multires	cage generation/interval, connectivity, hierarchy and level/detail correspondence when stateful	from validated state/checkpoint; topology event rebuild or fallback
Volumes	stateless	frame by frame

Table 6: Checkpoints every $N = 10$ frames, keeping the last two (≈ 3 GB).

Bound the immutable frame queue and apply backpressure. The end-to-end three-hour target for 240 frames permits 45 s per frame on average, including startup, pipeline fill/drain, all consumers and I/O. Stage medians alone cannot establish it. A candidate allocation is 32 s Simulate GPU, up to 2 s neural Mesh GPU and 6 s Whitewater/Volumes GPU per frame, overlapping up to 33 s CPU Mesh/Multires service with separately bounded I/O. These are allocations to test, not independent timing promises. Measure shared-GPU/CPU/storage contention, cold model load, initialization and queue drain; apply backpressure and do not assume Mesh stays off the critical path.

14.5 Protocol, settings and Blender delivery

The subprocess emits only versioned JSON Lines on stdout and structured logs on stderr, with a heartbeat at least 1 Hz reporting the last coherently known progress. A stdin `cancel` request stops at a safe boundary while retaining published output; timeout fallback uses platform-appropriate termination. All `bpy` calls stay on Blender's main thread, using timer polling and transient job state outside persistent Blender ID data.

Materialise defaults in persisted resolved scene/settings records and fingerprints (`ForceDefaults` or equivalent), so a later default cannot reinterpret an old scene or cache. Numeric field/enum IDs are stable and CI runs `flatc --conform`; Python dictionaries do not define schema evolution. Deliver versioned `.blend` assets for points, Surface LOD, Whitewater, foam projection, attribute colours and volumes, with water/foam and whitewater materials. Headless tests use the same public inputs as artists. Native cache readers must demonstrate render playback without a Python callback per frame.

15 Verification contracts and evidence boundaries

15.1 Validation

Unit oracles must be independent of the implementation: B-spline nesting masks and Greville reproduction, `patch_test_t_junction`, `mac_apic_conservation`, `apic_c_across_transition`, `work_identity`, `flip_gravity_free_fall`, `asflip_alpha_once`, `asflip_beta_gate`, `g2p_order_sentinel`, `elastic_null_mode`, `cpic_mask_identities`, `flip_residual_wall`, split/merge bounds, two-phase and two-layer flux with θ, static-drop jump orientation, moving-paddle flux, sealed-box compatibility, cut-cell divergence, SPD checks, constitutive invariants, sign tests (density correction, buoyancy, volume offset), and infrastructure tests (layout ABI, fixed-point order invariance, sort determinism, bitwise rollback).

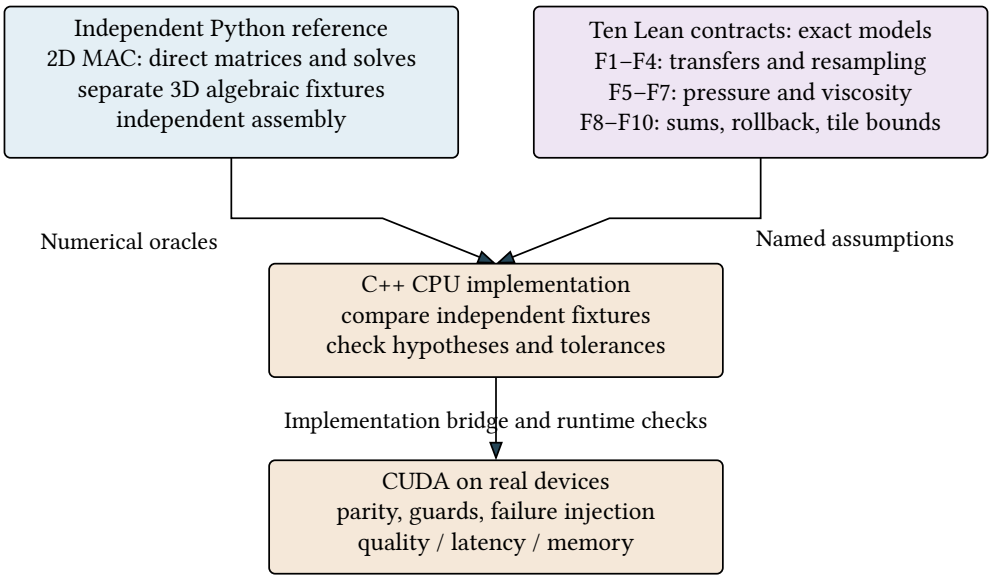
Functional tests cover the dam break and energy look-book, hydrostatic column, volume drift, piston work, Rayleigh drop, Couette and planar extension, wave relaxation, band crossing, hollow-shell and deep-tank surfaces, gas vortex ring, explosion and sealed heating, whitewater sub-step invariance and ballistic halo, stage invalidation, checkpoint resume and CPU/CUDA parity. Additional oracles cover masked constant/affine gathering, old/new ghost no-force FLIP, compressed reseed/delete/level cycles, constituent-volume merge, generalised viscosity eigenvalues on slivers and contrasts, the thin-slab continuous sweep, full acoustic directional differentiation, retained-support defect budgets, opposing whitewater bursts and stiff exact bubble drag. Reliability acceptance additionally requires 50 crash/cancel/device-error injection runs with every published frame readable, 1,000 cumulative default-scene frames without unresolved retries, nonfinite accepted state or warnings, deterministic interrupted/resumed output, and at most 1 % unexplained live-memory growth from frame 10 to 240 in the fixed job. Separate live allocations, reserved high-water arenas and expected scene/output growth. All supported examples/defaults must succeed before release; one successful scene is insufficient. A skipped test is not a passed test. Budgets are < 30 s unit, < 5 min functional, < 30 min end-to-end.

15.2 Independent numerical reference and formal contracts

A small independent float64 NumPy/SciPy reference is an acceptance prerequisite for optimising each corresponding numerical feature. Its uniform 2D MAC model directly assembles tiny pressure and viscosity matrices and uses a direct sparse solve. It covers transfer, closed and free-surface pressure, moving-wall flux, source compatibility, weak compliance, explicit and implicit viscosity, and particle/finite-volume ownership. Independent 3D algebraic fixtures cover staggered angular momentum and cubic MAC split. Export versioned intermediate arrays with units, state versions, residual conventions and tolerances for C++ CPU and CUDA differential checks. Python does not implement Blender, production sparse allocation, a mesher or a second full simulator.

The production CPU backend tests the actual layouts, schema, I/O and computational path; agreement with CUDA alone cannot detect a shared assembly error. Python and production must therefore have independent assembly implementations. A fixture must exercise a stated hypothesis and include a counterexample that fails when that hypothesis is removed.

The initial Lean 4 scope is the following ten finite-dimensional theorem families [34], using finite index types, real vectors and matrices, integer reductions and a pure transition model. They are stage commitments, not interchangeable optional pilot suggestions. The complete obligations and numerical bridges are stated in the accompanying formal-contract reference this section records their scientific scope.



Four separate results: compiled contracts · independent fixtures · implementation checks · device measurements.

Exact proofs do not certify IEEE arithmetic, C++ memory safety, CUDA races, geometry or visual quality.

Figure 10: Independent assembly, exact contracts, production parity and target-device evidence answer different questions. None stands in for the others.

15.2.1 Transfer and resampling contracts

F1: reconstruction. For a complete row stencil require $\sum_i w_i = 1$, $\sum_i w_i r_i = 0$, invertible $D_a = \sum_i w_i r_i r_i^T$, and derivative moments $\sum_i \nabla w_i = 0$, $\sum_i r^T_{i(\nabla w_i)} = I$. Substitution of $u_{a(x_i)} = v_a + c_a r_i$ proves the gather, affine fit and derivative separately reproduce their affine values. Identical old/new arrays give zero grid increment; eliminating the particle FLIP residual additionally requires its particle value to equal the gathered old field. Ghost values must equal the same compatible affine continuation. Renormalisation of a masked stencil establishes partition, not its original first moment; the theorem does not supply missing ghost completeness.

The uniform F1 deliverable also defines the actual quadratic/cubic weights and proves their piecewise-polynomial identities; merely assuming partition and moments is not that complete target. THB remains subject to explicit hypothesis checks.

F2: unmasked MAC deposition. Complete unmasked first-moment stencils conserve mass and linear momentum per component. For physical mass deposition the three staggered arrays represent the same mass, not three additive masses. In the Gas marker branch the corresponding theorem concerns numerical deposition weight, with physical species inventories checked separately. For components j, k, the discrete angular momentum is

$$\sum_{\text{faces } k} x_j q_k - \sum_{\text{faces } j} x_k q_j = \sum_p m_p [x_{pj} v_{pk} - x_{pk} v_{pj} + c_{pk} D_{pk} e_j - c_{pj} D_{pj} e_k]. \tag{48}$$

Expand $x_i = x_p + r_i$ and use the row-specific second moments. The claim excludes force, contact, remap and quantisation; cut/ghost budgets are separate.

F3: symmetric split. Symmetric offsets have zero mean and covariance S. With $R_a = D_{\text{parent},a} - D_{\text{child},a} - S$, direct expansion gives child energy equal to parent energy minus $(\frac{m}{2}) \sum_a c_a R_a c_a^T$. If R_a is supported only along coordinate a, angular momentum remains exact. Quadratic isotropic splitting has zero deficit. Cubic-normal/quadratic-transverse MAC instead has $S = 3h^2 \frac{I}{16}$, $R_a = h^2 e_a \frac{e_a^T}{16}$ and exact loss $mh^2 \sum_a \frac{C_{aa}^2}{32}$. The theorem assumes uniform unmasked admissible supports; it does not prove that children avoid obstacles.

F4: merge. Positive masses and positive-definite row moments define K_a, H_a and the target affine row as in the merge equations. The gate $MD_a^* - K_a \geq 0$ for every row bounds affine energy by the child energy through the block-Gram inequality and inverse order. First and antisymmetric moments prove linear and angular momentum; additive reference/current volume and current-volume-weighted concentrations preserve constituent volumes. A floating eigenvalue tolerance must refuse uncertain gates rather than treating exact PSD as a numerical guarantee.

15.2.2 Pressure and dissipation contracts

F5: pressure quadratic form and kernel. For integrated incidence R, positive face conductance k, and nonnegative Dirichlet/compliance diagonals d, c,

$$A = R\ \mathrm{diag}(k) R^T + \mathrm{diag}(d + c), \quad p^T A p = \sum_f k_{f(R^T p)_f^2} + \sum_{i(d_i + c_i)} p_i^2. \tag{49}$$

Every term is nonnegative. The kernel consists of constants on each unanchored positive-conductance component, vanishing on anchored components. This gives SPD after the correct gauges and solvability exactly when the right-hand side sums to zero on each unanchored component. A positivity lemma alone does not prove the kernel or compatibility conclusions.

F6: correction residual. With consistent incidence, jumps, wall flux and boundary lifting, take $b = S - RF^\star - W + Cp_{\text{old}}$, $A = L + C$ and $RF_{\text{new}} = RF^\star + Lp$. For $r = b - Ap$, substitution gives

$$RF_{\text{new}} + W - S + C(p - p_{\text{old}}) = -r. \tag{50}$$

The identical coefficients must appear in assembly and correction. Positive cell-volume division yields the physical stopping criterion; finite compliance includes its volume-change term, so nonzero divergence alone is not a failed solve.

F7: diffusion energy. With positive diagonal M, $K = E^T W E \geq 0$ and a certified $KM^{-1}K \preceq \Lambda K$, explicit Euler is nonexpansive in $E_{M(u)} = u^T M \frac{u}{2}$ when $\Delta t \Lambda \leq 2$. Expansion gives the energy identity in § 7.1; the selected Gershgorin-to-spectral bridge is itself a proof obligation. Backward Euler with wall forcing satisfies

$$E_{M(u^+)} - E_{M(u)} = \Delta t (u^+)^T b_{\text{wall}} - \Delta t (u^+)^T K u^+ - \left(\frac{1}{2}\right) \|u^+ - u\|_M^2. \tag{51}$$

This proves dissipation relative to wall work; it makes no absolute contraction claim for a moving wall.

15.2.3 Execution contracts and acceptance gates

F8: integer deposition. For $N \le 2^n$, $0 \le n \le 62$, $s > 0$, finite c_j and nearest-rounded $Q_j = \operatorname{rint}\left(\frac{c_j}{s}\right)$ with $|\frac{c_j}{s}| \le 2^{62-n}$, every partial sum has absolute value at most $\sum|Q_j| \le 2^{62}$. Integer commutativity gives order invariance, and $|s\sum Q_j - \sum c_j| \le N\frac{s}{2}$. State ties-to-even explicitly. The theorem does not establish identical floating contributions on CPU and GPU, or safe conversion of NaN.

F9: transactions. A pure state transition preserves immutable committed state on rejection; induction over rejected attempts preserves particles, grid, topology, pressure history, physical source inventory, accepted events, IDs/RNG and accepted time. One acceptance swaps once and adds its accepted Δt once. Attempt diagnostics and the smaller retry step are separate outputs. Adaptation is separately transactional. Deterministic replay assumes a deterministic trial function; it does not prove CUDA race freedom or C++ nonaliasing.

F10: tile inclusion. The half-offset bin implies $X_0 \in [4B + 1.5, 4B + 5.5)$; with $|X - X_0| \le 0.5$, $X \in [4B + 1, 4B + 6)$. Floor inequalities place the three quadratic samples for shifts zero and one-half, and the four cubic-normal samples, in $[4B, 4B + 8)$. Tensor products give the uniform 8^d tile. This assumes the displacement bound; it proves neither its estimator nor THB support inclusion. Runtime inclusion checks remain mandatory.

Gate	Formal targets	Independent numerical bridge
M0	Pinned Lean/mathlib project; F5 quadratic-form and F8 range/order pilots; measured proof effort.	Tiny directly assembled pressure matrices and integer adversarial fixtures.
M1	F1 uniform basis; F2, F5, F6, F8, F9, F10.	Transfer and direct projection, quantised reductions, failure-injection rollback and allocation guards.
M3	F1 cubic/transition hypothesis checks; F3 and F4.	3D split/merge moments, patch tests and refused masked supports.
M4	F7; exercise F6 finite-compliance cases.	Small-mass/contrast viscosity, wall work and compliant physical residual.

Table 7: Feature claims depend on their theorem families and independent implementation bridges.

The two M0 pilots establish project feasibility and proof effort, not permission to replace the ten-contract scope by two lemmas. Public targets must compile without `sorry`, `admit` or unapproved project axioms; retain `#print axioms` output and allow only the recorded foundations of the pinned environment. F1 assumes moment identities for a selected transition basis: numerical THB patch tests still have to establish that the implementation satisfies them. A noncritical deferral requires an explicit ADR naming the missing theorem, reason, numerical substitute and revisit milestone.

Report four distinct evidence classes: kernel-checked exact-arithmetic contracts, independent numerical-reference results, C++/CUDA parity and runtime invariants, and real-device end-to-end quality/performance. None replaces another. The contracts do not prove Navier–Stokes, total simulation stability, IEEE bounds, memory safety, collision geometry or reconstructed topology. No reference solver, Lean project or checked theorem is claimed implemented here.

15.3 Product constraints and measured choices

Product decisions are recorded:

- **Scene**: sloping beach, Stokes swell, fixed camera (§ 1). **D1**: sheets below $\Delta x_{\min}$ are carried by the mesh only. **D2**: `sim/` holds band particles every frame and all particles at checkpoints.
- **D3**: one MAC grid for all continua. **D4**: foam projected in Geometry Nodes; Mesh never reads `whitewater/`. **D5**: physical low-Mach gas with few parameters.
- **D6**: fractional viscosity step, coupled Stokes only if the planar-extension test requires it. **D7**: two-way rigid bodies at M6. **D8**: Ryzen 7 9700X, 64 GB, 3 TB.
- **D9**: adaptivity is a firm objective; at most six hours is tolerated temporarily without it, and an AG-MPM/GIMP alternative is tried before any abandonment. Supersonic flow is not implemented but kept addable. **D10**: unresolved trapped air may collapse before M4; resolved cavities receive physical pressure at M4. Finite-c liquid is a design option with physical grid pressure history.

What remains is settled by measurement, not by product decision: the exact flow-map gate thresholds (§ 10.4), validation of weak liquid compressibility (§ 10.3, M4), and builder values.

15.4 Evidence boundaries

No FluidiTy simulation result is reported here; source-inspection facts and cited platform timings are explicitly attributed. The design equations are contracts, and the budgets (three hours, or six without adaptivity; 11 GB; the 4.8 GB estimate) are acceptance objectives and estimates, not CUDA results. The largest risks are a 3D THB-MPM MAC basis on GPU (not validated by the cited sources; go/no-go after the uniform baseline, with the D9 fallback), a tracked mesh combined with multiresolution (integration unvalidated), memory bandwidth, the cost of fixed-point P2G, and reading caches in Blender without Python. Validation proceeds milestone by milestone with exit criteria measured on the target GPU. The storage schema, optional modules and memory examples are engineering choices of this project; they should not be attributed as a single published algorithm to any reference below.

Detailed substep · ①–⑦: emission, activation, P2G, forces and contact

Gray: shared operations; purple: MPM solids; blue: fluids; orange: gas; green: control. Each component a has its own face lattice. All support positions are frozen at x^n; the accepted state is A, the trial state is B.

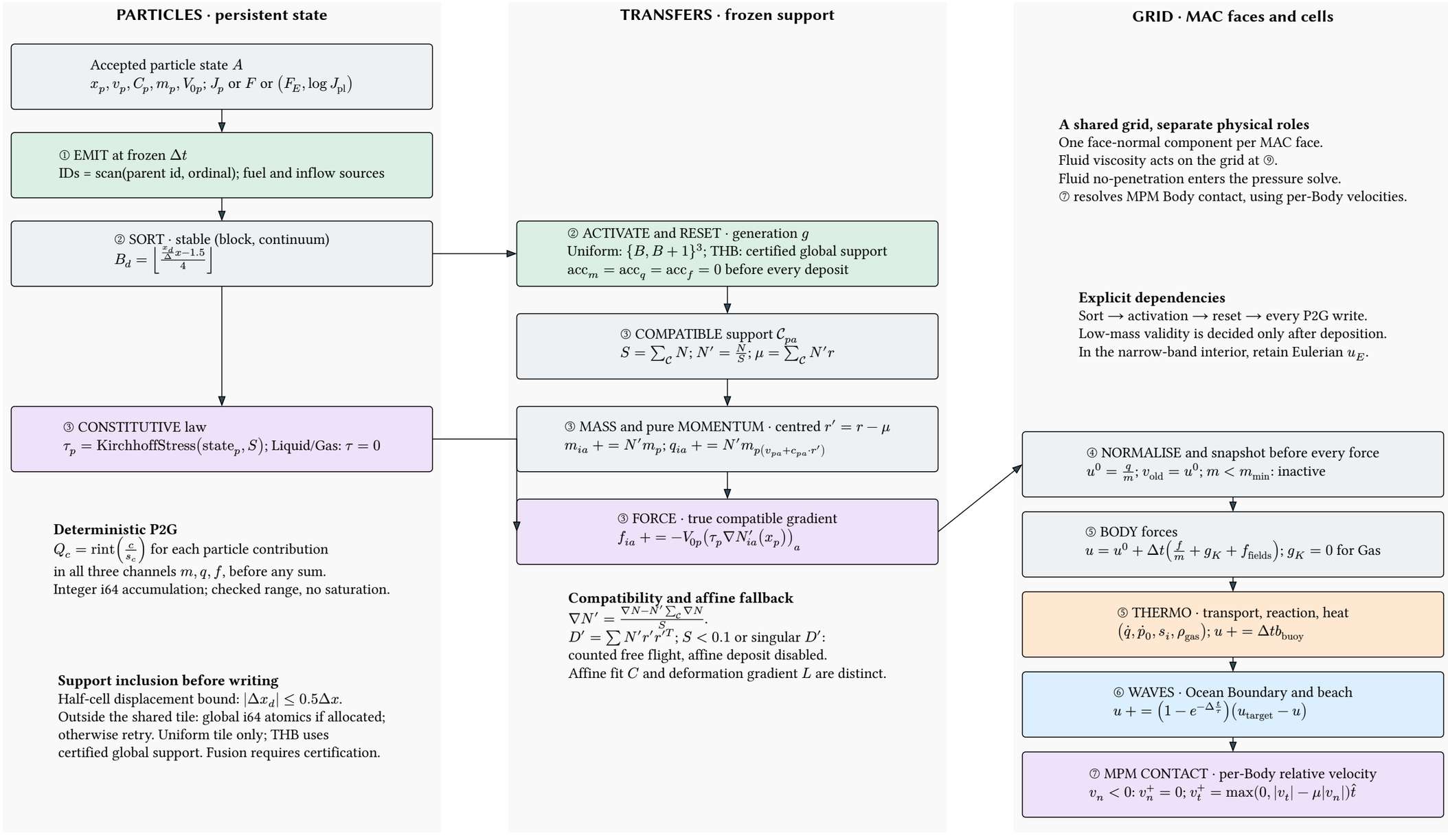


Figure 11a. Steps ①–⑦. The FLIP snapshot precedes application of stress forces, gravity, buoyancy, waves and contact. Compatible P2G uses centred moments; activation and accumulator reset are prerequisites of deposition. Disabled operators are identities.

Detailed substep · ⑧–⑩: geometry, dissipative viscosity and pressure

Interface geometry is built before viscosity. Pressure is the last physical operation on grid velocity before G2P; extrapolated ghost values are then built for both the old snapshot and the new velocity.

PARTICLES · persistent state

Current volume → grid inertia
$V_p = J_p V_{0p}$; $\rho_p = \frac{m_p}{V_p}$.
Face inertia: $M_f = a_f \frac{d_f}{\beta_f}$.
The reference volume enters solid stress forces.

A single viscous operator
Tensorial strain Eu includes cross derivatives.
$u^T K_\mu u \geq 0$: homogeneous boundaries dissipate in both branches under the explicit certificate.
Moving walls add physical work through b_{wall}.
Particle C carries no viscous stress.

Finite sound speed
$\tilde{p}$ is physical pressure where $c < \infty$.
$p^\star = \text{Advect}(\tilde{p}^n)$ on all fluid cells.
$K = \rho_0 c_{\text{ref}}^2 = \rho c_{\text{eff}}^2$; commit $\tilde{p}^{n+1} = p$ on acceptance.
Compliance retains the mean: no gauge, no incompressible compatibility test there.

Pressure validation in ⑬
$r = b - (\mathcal{L} + H)p$; $H = \text{diag}(h_i)$.
$\text{div}\, u_i - s_i = -\frac{p_i - p_i^\star}{K_i \Delta t} - \frac{r_i}{V_i}$.
Require this compliant residual to converge.
At $c = \infty$, recover $\text{div}\, u = s$.

TRANSFERS · frozen support

Face conductance
$k_f = \Delta t a_f \frac{\beta_f}{d_f}$.
Two phases: $\beta_f = \frac{1}{\theta \rho_i + (1-\theta)\rho_j}$.
$b_i = V_i s_i - Q_i^\star - Q_{\text{wall},i} + h_i p_i^\star + \text{jumps}$.
Free surface: $p_\Gamma = p_{\text{gas}} + \sigma\kappa$, $J_f = 0$ at distance $\max(\theta, 0.05) d_f$.

Conservative geometry and work
Coarse flux = sum of fine physical fluxes.
$G_{\text{int}} = -D^T$; pressure force $= -G_{\text{int}} p$.
No volume division of matrix rows.
The same a_f, β_f, d_f, J_f enter solve and correction.

Explicit/implicit selection
$S = M^{-\frac{1}{2}} K_\mu M^{-\frac{1}{2}}$.
$\Lambda_{\text{cert}} = \max_i \sum_j |S_{ij}| \geq \lambda_{\max(S)}$.
Cut cells, varying density/viscosity and THB are included in that bound.
Without a certificate, use the implicit solve.

Ghost construction
Incompatible or inactive faces use extrapolated data.
Replace their blade-normal velocity by blade velocity, consistently for old and new fields.
A constant boundary velocity gives zero FLIP increment.

GRID · MAC faces and cells

⑧ INTERFACE at x^n · one plane per cell
`sdf_sim` → fractions, apertures $a_f, \theta, \kappa, \beta_f, J_f, s_{\text{corr}}$

⑨ VISCOSITY · all fluid faces, band and interior
$K_\mu = E^T W_\mu E$; $W_\mu = 2\mu V$; $K_\mu \succcurlyeq 0$

⑨ EXPLICIT only with a certified spectral bound
$\Delta t \Lambda_{\text{cert}} \leq 2(1-\varepsilon)$
$u = u^\star + \Delta t M^{-1}(b_{\text{wall}} - K_\mu u^\star)$

⑨ otherwise IMPLICIT
$(M + \Delta t K_\mu) u = M u^\star + \Delta t b_{\text{wall}}$

⑩ COMPATIBILITY · sealed incompressible components
$\sum V_i s_i = \sum_{\text{walls}} (|f| - a_f) u_{\text{obst}} \cdot n$; fix one gauge

⑩ SPD SOLVE · integrated rows, MG-PCG
$\sum_f k_{f(p_i - p_j)} + h_i p_i = b_i$; $h_i = \frac{V_i}{K_i \Delta t}$

⑩ CORRECT · identical coefficients and jumps
$u_f = u_f^\star - \left(\Delta t \frac{\beta_f}{d_f}\right)(p_j - p_i - J_f)$

⑩ → ⑪ GHOST extension on full frozen support
$(v_{\text{old}}, u) \to (\hat{u}^{\text{old}}, \hat{u}^{\text{new}})$; apply blade-normal data

Figure 11b. Steps ⑧–⑩. The viscosity boxes denote alternative branches. Finite-c cells retain their pressure as physical state and are validated against the compliant continuity equation. The pressure interface jump J_f is distinct from particle volume ratio J_p.

Detailed substep · ⑪–⑭: full-stencil G2P, collision and acceptance

G2P reads the full basis $N, \nabla N$ at frozen x^n, including old and new ghost values. Each proposed write belongs to trial state B until ⑬ accepts it; scheduled adaptation is a separate transaction afterward.

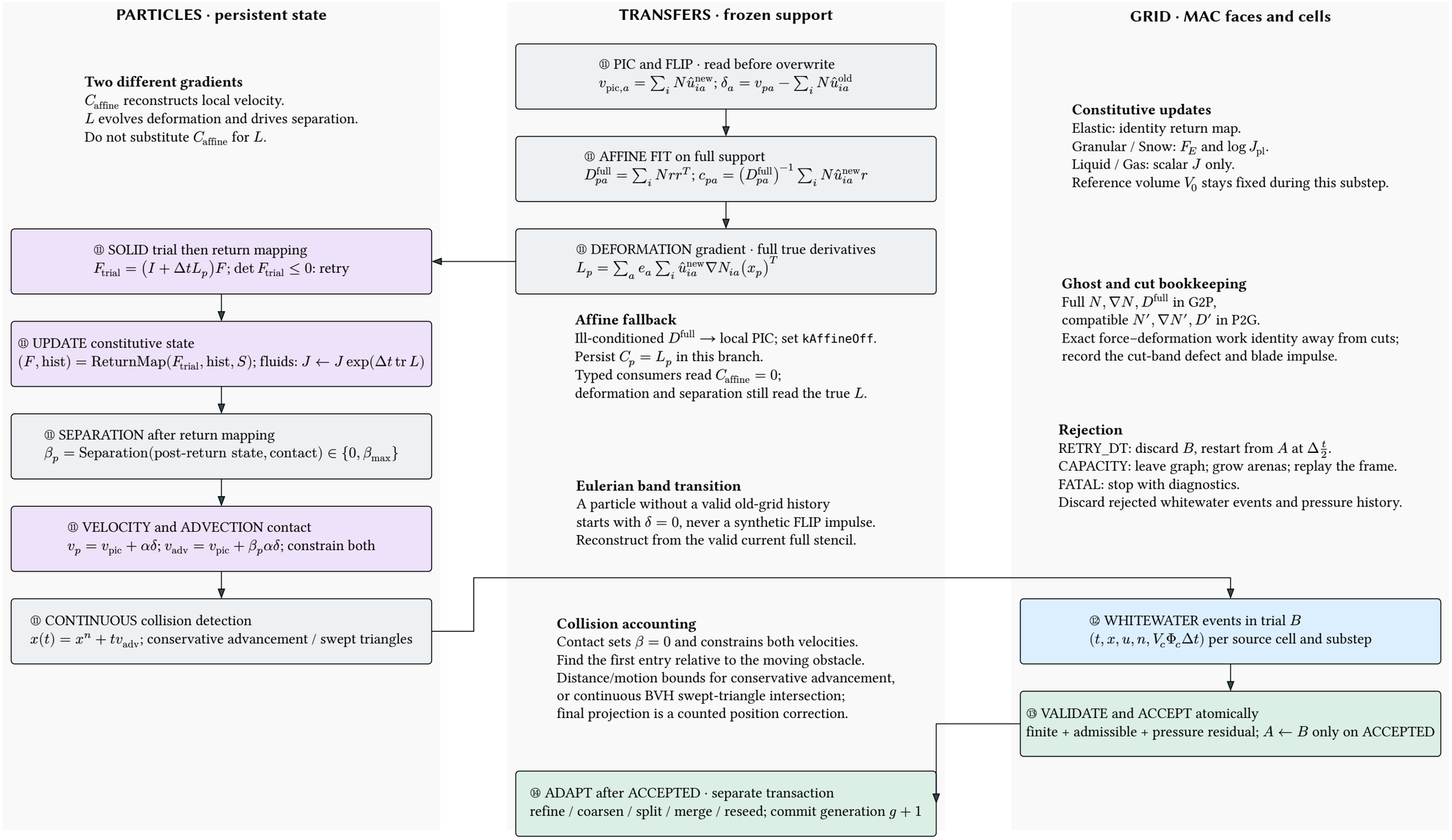


Figure 11c. Steps ⑪–⑭. α enters the FLIP contribution once. Contact constrains the actual advection velocity, and continuous collision detection prevents tunnelling. Whitewater events are committed with the accepted physical substep; adaptation cannot invalidate that accepted state.

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